iView ETG

User Manual

Version 1.1

12.04.2012

SMI
SensoMotoric Instruments

Copyright

iViewETG is owned by SensoMotoric Instruments GmbH or its suppliers and is protected by the Federal Republic of Germany copyright laws and international treaty provisions. Therefore, you must treat the SOFTWARE like any other copyrighted material (e.g. book or musical recording) except that you may either
a) make one copy of the SOFTWARE solely for backup or archival purposes or
b) transfer the software to a single hard disk provided you keep the original solely for backup or archival purposes.
You may not copy the written materials accompanying the SOFTWARE.
The user is not entitled to allow a third party to use the software simultaneously without written approval of SensoMotoric Instruments GmbH. Independent branch offices or subsidiary companies are also understood to be a third party in this sense. SensoMotoric Instruments GmbH and/or its supplying firm remain the owners of the delivered software, even if it is altered.

Table of Contents

Part I Introduction 2

1 How to Read this Document ... 3

2 General Advice ... 4

3 Maintenance ... 7

4 Declaration of Conformity ... 8

Part II Product Overview 12

1 Description ... 13

SMI-ETG ... 13

SMI-ETG Laptop ... 16

2 Package content ... 16

3 Accompanying Software ... 18

4 Technical Specifications ... 19

Part III Quick Start 24

1 Recording Process ... 24

2 Quick Start Guide ... 25

Doing a Quick Run ... 25

Run Experiment ... 26

Part IV SMI Eyetracking Glasses Hardware 30

1 Hardware Setup ... 30

(c) 2012 SensoMotoric Instruments

Nose Rest ... 32
Convertible Eye Glasses ... 33

Part V iViewETG Software 36

1 Starting iViewETG Software ... 36

2 Preparing a Quick Run ... 37

Quick Run Settings ... 37
Default Calibration Mode ... 39
Allow Different Calibration Mode ... 41
Eye Tracking Mode ... 41
Scene Camera Mode ... 41
Scene Camera White Balance ... 41
Microphone ... 42
Starting a Quick Run ... 42
Calibration ... 43
Quick Run Recording ... 44

3 Preparing Experiment Run ... 45

Handle Experiments ... 45
Creating a New Experiment ... 46
Opening an Existing Experiment ... 47
Close Experiment ... 48
Experiment Run Settings ... 49
Lock/Unlock Settings ... 49
Participant Properties ... 50
Experiment Settings ... 53
Hardware Configuration ... 55
Starting an Experiment Run ... 57
Calibration ... 59
Experiment Run Recording ... 59

Part VI In-Depth Information 62

1 New Respondent ... 62

2 Calibration ... 64

0-Point Calibration ... 65

1-Point Calibration ... 67
3-Point Calibration ... 68
Calibration Reset ... 71

3 Gaze Cursor Stabilization ... 72

Eye Position ... 73
Monocular Vision ... 73
Make Up ... 73
Ocular Dominance ... 74
Distracting IR Reflection ... 74
Corrective Glasses ... 74
Contact Lenses ... 74

4 User Interface ... 75

Dashboard ... 75
Settings Screen ... 76
Recording Screen ... 77
Recording Display ... 78
Trackability ... 78
Recording Time ... 79
Tracking Ratio ... 80
Calibration ... 80
Microphone ... 81
3-Step-Confirmation ... 82
Runtime Information ... 83

5 Data Storage Structure ... 83

Change Data Storage Directory ... 84
Directory Structure ... 86
IDF Files ... 87
Video Files ... 87

6 Software Updates ... 88

Part VII Appendix 90

1 Troubleshooting ... 90

2 Third Party Licenses ... 90

(c) 2012 SensoMotoric Instruments

GNU Lesser General Public License 91
Boost Software License 95
BSD License 96
GNU General Public License version 3 97
GCC Runtime Library Exception 112

Part VIII License Agreement and Warranty 116

Part IX About SMI 124

Index 127

Introduction

Chapter



1 Introduction

Congratulations on purchasing the SMI Eye Tracking Glasses, a mobile gaze tracking device by SensoMotoric Instruments, and the accompanying iViewETG software for experimental design and recording. The SMI-ETG support a broad range of mobile eye tracking studies, e.g. the evaluation of brand awareness at the point of sale, of user experience on mobile devices, of man-machine interaction or of training methods in professional sports. SMI-ETG are as fast to set up and as easy to use as a pair of normal glasses

Please read this manual carefully to ensure best results when working with the SMI-ETG.

You can find more information on the SMI-ETG as well as sample recordings at: www.eyetracking-glasses.com



SMI-ETG - mobile eye tracking device

(c) 2012 SensoMotoric Instruments

1.1 How to Read this Document

This manual is designed to serve both as online help and as printed documentation of SMI Eye Tracking Glasses (ETG), the mobile gaze tracking device and its operating software iViewETG for experimental design and recording. The software version covered in this document is: Value Candidates

You can use this manual in either one of these ways:

Read through the chapters pertaining to particular functions to get background information before using the hardware and the software program. Consult the manual as a reference document to find out particular information. You can find a topic either by consulting the table of contents (at the front of the manual), or the index (at the end).

The information in this manual can also be accessed through the iViewETG software program. Press [F1] to open the Online Help on a menu item or on the element that has currently the input focus or that is selected.

It is probably not necessary to read all the chapters consecutively as every chapter is complete within itself. This means that some phrases may reoccur. You may scan through the chapters to look for the topics that interest you. The link references included in the text as well as the table of contents and the index should help you find your way through this document. You can locate information quickly by using the Online Help's table of contents, index or full text search features.

Nomenclature:

- SMI: SensoMotoric Instruments GmbH
- SMI-ETG: Eyetracking glasses device
- SMI-ETG laptop: laptop for use with SMI-ETG
- iViewETG: Eyetracking glasses operating software

- BeGaze: Eyetracking data analysis software
- Link: please follow the given link to get more detailed information about this topic

1.2 General Advice

 **Please read carefully before using the SMI-ETG!**

- The SMI-ETG is a sophisticated measurement device.
- Handle with care!
- Always use the SMI-ETG straps to ensure firm fit of the glasses on the participant's head!

 (c) 2012 SensoMotoric Instruments

- Do not fold the glasses!



- Do not over bend the glasses arms! Extending the temple arms beyond the mechanical stops will break the hinges.



- SMI-ETG does not provide special eye protection! It does not comply with any mechanical or optical safety class for external light sources!
- Do not scratch the convertible eye glasses!
- Operation and storage conditions: temperatures between 5°C and 40°C and a humidity not exceeding 80%!
- Frequently check SMI website for free Software updates: www.smivision.com/software
- When NOT in use, the SMI ETG should be unplugged from the USB port of the ETG-Laptop.
- Do not unplug the device from USB port when in use.
- Check the technical safety of the device, e.g. check for damage to the housing etc. periodically.
- Only use original components (cables, power supply etc.).
- In case a component needs to be replaced, only replace it with original components provided by SMI.
- Do not attempt to repair the system by yourself.
- Unplug the Eye Tracking Glasses from the ETG-Laptop before cleaning.
- No part of the product may be modified or rebuilt.
- The system must not be used in presence of explosive or flammable gases.
- Any usage other than described in this manual is not permitted.
- Never operate ETG-Laptop without battery except when hot swapping the battery!
- Do not connect the SMI-ETG ETG-Laptop to the internet!



SMI does not assume liability for resultant damages to property or personal injury if the product has been misused in any way or damaged by improper use or failure to observe these operating instructions. In addition, any unauthorized modifications or repairs of the device will render the warranty null and void!



Make sure the presented visual stimuli or the environment in which you conduct your study do not harm or injure your

 (c) 2012 SensoMotoric Instruments

participants. SMI is in no way responsible for the experiments you develop, execute, and analyze. Do not offend your participant's cultural background, age or psychological condition.

1.3 Maintenance

Do not attempt to repair the system by yourself.

- Servicing, adjustment or repair works may only be carried out by a certified distributor or SensoMotoric Instruments GmbH (SMI).
- In case your SMI-ETG or a component of the system is damaged, unplug SMI-ETG from the ETG-Laptop. Do not use the SMI-ETG until the system or its component has been repaired by a certified distributor or the manufacturer SensoMotoric Instruments GmbH (SMI).
- To clean the convertible eye glasses of the SMI-ETG, remove them from the SMI-ETG frame and clean with water or using the supplied cleaning cloth.
- To clean the SMI-ETG frame, use a dry towel. Do not use water. Make sure that camera modules are not exposed to any liquids.

1.4 Declaration of Conformity

Declaration of Conformity

SensoMotoric Instruments GmbH

Address Warthestr. 21
14513 Teltow
Germany
Phone +49 (3328) 3955 0
Fax +49 (3328) 3955 99
Web www.smivision.com
Email info@smi.de

SensoMotoric Instruments GmbH hereby declares that the following product

Type of Product Eye Tracking Equipment
Product Name SMI Eye Tracking Glasses (ETG)
Model Designation ETG (Glasses)
Product Description Mobile Eye Tracking Device

is in accordance with the following European directives

2004/108/EEC Electromagnetic Compatibility (EMC)
2006/95/EEC Low Voltage Electrical Equipment
93/68/EWG CE marking

and complies with the following European standards:

EN 61010-1:2002 Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 1: General requirements (IEC 61010-1:2001)
EN 61000-6-3:2007 Electromagnetic compatibility (EMC) - Part 6-3: Generic standards - Emission standard for residential, commercial and light-industrial environments
EN 62471:2009 Photobiological safety of lamps and lamp systems

Teltow, October 20th, 2011

Eberhard Schmidt
Managing Director
SensoMotoric Instruments GmbH

SMI-ETG comply with the following European directives:

- 2004/108/EEC - Electromagnetic Compatibility (EMC)
- 2006/95/EEC - Low Voltage Electrical Equipment
- 93/68/EWG - CE marking

SMI-ETG comply with the following European standards

- EN 61010-1:2002
- EN 61000-6-3:2007
- EN 62471:2009:

Product Overview

Chapter



2 Product Overview

SMI-ETG human interface is a non-invasive video based glasses-type eye tracker with integrated audio recording.
The device is worn like a normal pair of glasses and includes a HD scene camera and special eye tracking technology which captures the eye movements of the person wearing it.

- SMI-ETG are delivered with a ETG-Laptop for mobile recording and live view options.

For further technical details please refer to Technical Specifications 19 and Hardware Components. 30

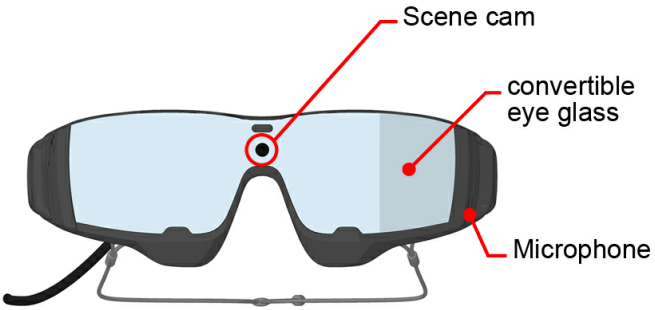
The SMI-ETG delivery contains the following parts:

(c) 2012 SensoMotoric Instruments

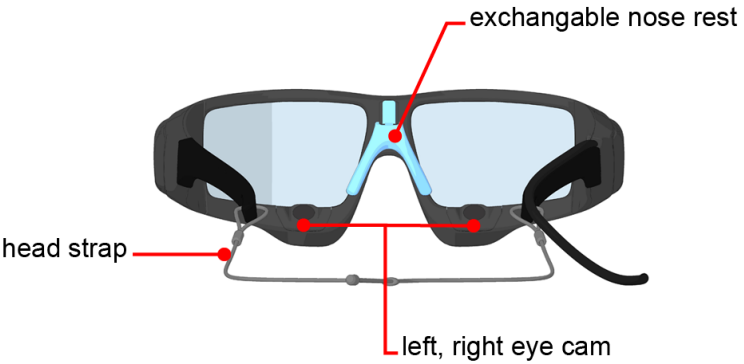
2.1 Description

2.1.1 SMI-ETG

Front View:

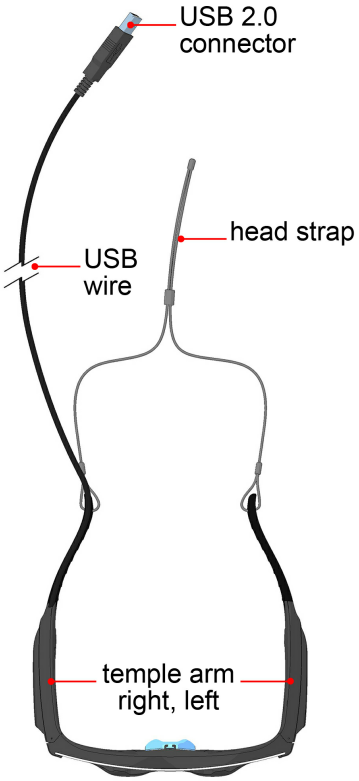


Back View:

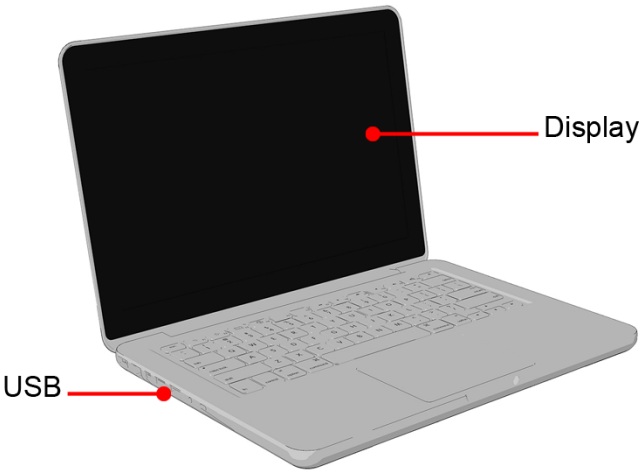


(c) 2012 SensoMotoric Instruments

View from the top:



2.1.2 SMI-ETG Laptop



2.2 Package content

Please make sure the following items have been delivered:

1) ETG-Laptop
2) Power supply for ETG-Laptop
3) SMI-ETG with attached head-strap
4) Additional nose rests (for adaptation of eye position 73)
5) Transparent infra-red filtering eye glasses (inserted in the SMI-ETG) absorbing distracting IR reflection 74
6) Shaded, infra-red filtering eye glasses absorbing distracting IR reflection 74
7) License dongle for SMI BeGaze Analysis Software (level of

functionality depends on purchased license)

Additionally there is an empty compartment that can be used to transport a spare battery for the ETG-Laptop (9). A spare battery is not included in the SMI delivery. Spare batteries can be purchased from the manufacturer of the SMI-ETG laptop.

All components are delivered in a special flight case to protect the SMI-ETG and laptop from damage.



Flight case with ETG-Laptop placed on top of SMI-ETG



Flight case with SMI-ETG and additional hardware components

2.3 Accompanying Software

- iViewETG: Controller software for experimental design and recording of eye tracking data.

- SMI Experiment Suite 360°: **SMI BeGaze Mobile Video Viewer** software for qualitative video and eye tracking data analysis. This software allows the experimenter to view and export scan path videos of individual participants, to view and export scene videos with gaze cursor overlay of individual participants, to export raw gaze data and gaze event data, and to perform off-line calibrations.

For further information about iViewETG please refer to iViewETG Software 36.

For more information about SMI BeGaze Analysis Software please refer to the SMI BeGaze manual.

For more information about license options and additional packages please contact your SMI sales representative.

2.4 Technical Specifications

Human interface design	Non-invasive video based glasses-type eye tracker with integrated audio recording
Glasses	Size:168mm (B) x 58mm (H) x 173mm (T) (6.6” (W) x 2.3” (H) x 6.8“ (D)) Weight: 75g (0.17lbs) Estimated age compatibility (?): 7+ Head width (ear-to-ear): 138 -180 mm (5.4 - 7.1“)
Accessories	Convertible eye glasses, shaded and non-shaded
Calibration	Instant gaze cursor with 0-point,1-point and 3-point calibration mode (pre-recording), as well as offline calibration
Eye tracking principle	Binocular with automatic parallax compensation; dark pupil tracking
Temporal resolution	30Hz binocular
Spatial resolution	0.1°

Gaze position accuracy	0.5°
Tracking distance	40 cm - ∞ (15.8“ - ∞)
Gaze tracking range	80° horizontal, 60° vertical
HD scene camera	Resolution: 1280x960p with 24 fps; Video format: H264 Field of view: 60° horizontal, 46° vertical
Eyewear compatibility	Works with contact lenses and most selected corrective glasses
Recording time	ETG-Laptop: 2h, recording unit*: available at release
Recording devices	ETG-Laptop, recording unit*
	Interfaces: Network connection (Ethernet&WLAN)
	Application Programming Interface & SDK*: Online interfacing via SDK, remote wireless access and monitoring
	Video data: Online scene video with gaze position Numerical data: pupil radius, gaze base point (left / right), gaze direction (left / right), point of regard (binoc),
Norm compliance	CE Declaration of Conformity; EN61010-1:2001; Photobiological safety of lamps and lamp systems EN 62471:2009

 (c) 2012 SensoMotoric Instruments

iViewETG limitations See “add properties” 51	Max. number of participant Properties: 100 Max. number of Value Candidates: 40 Max. number of letters per Value Candidates: 256 Max. recording time per video: 2h
SMI BeGaze analysis limitations	Max. experiment recording time (sum of all recording times for all participants): 21h
SMI-ETG ETG-Laptop limitations	Max. recording time on battery power: 2h. This is only valid for batteries supplied by SMI. It is dependent on the individual condition of each battery as well. Max. recording time on AC-power: limited by free hard disk space available and experiment limiting factors (see limitations from iViewETG)

Quick Start

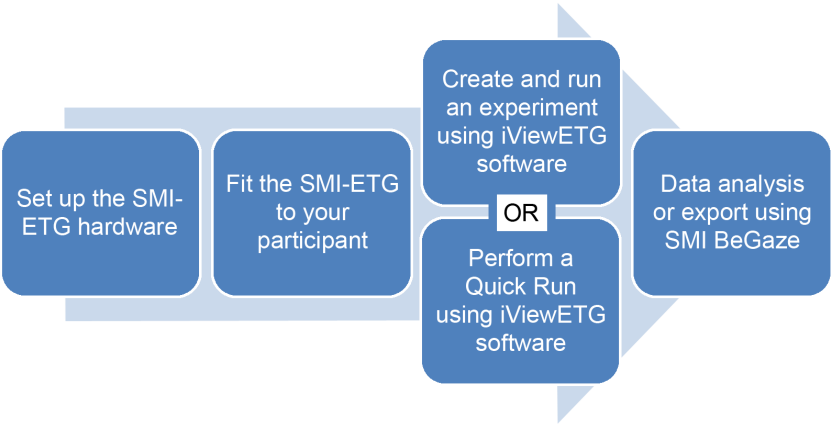
Chapter



3 Quick Start

3.1 Recording Process

In order to prepare and conduct your study with SMI-ETG and iViewETG, please proceed with the following steps:



(c) 2012 SensoMotoric Instruments

This process has two options for managing recordings. You can either do a Quick Run or Experiment Run:

- Quick Run 25: standard settings are used, data is stored separately for every participant (Data Storage Structure 83).
- Experiment Run 26: customized settings are used, data is stored in one experiment folder for all participants (Data Storage Structure 83).

3.2 Quick Start Guide

3.2.1 Doing a Quick Run

1. Fit the SMI-ETG to the participant's head (Hardware Setup 30).
2. Make sure the participant is wearing the SMI-ETG correctly. Please tighten the head strap to ensure a firm fit.
3. Launch iViewETG software.
4. Check whether the “Connected” symbol appears in the upper right corner of the screen. This indicates that the SMI-ETG are properly connected and ready to go. In case of other symbols appearing please see Hardware Setup 30.
5. Press "Quick Run" on the iViewETG dashboard. The Recording Screen 78 will appear.
6. Wait for the scene video to appear on the screen. This may take up to 30 seconds.
7. Press "New Respondent" to enter the name of the participant.

Before starting the calibration process, make sure that at least 30 seconds of time have passed since the participant has set up the SMI-ETG. This will assure the internal eye model has been completely adapted.

8. Perform a 1-point calibration:
   8.1 Press “1-Point-Calibration”.
   8.2 Ask your participant to look at a noticeable object in his field of view at a distance of 1.5m while keeping its head quiet.
   8.3 Move mouse over scene video view. Move the crosshair to the object where the participant looks at and click on the object (1-Point Calibration 40).

The system is now calibrated. You can check for calibration status in the Recording Display 78: The calibration status shows “1-Point Calibrated”.

9. Press "Record" to start the recording.

10. To stop the recording press "Record" again. When the red dot stops pulsating, the recording stops. The recorded data is getting saved automatically.

See Data Storage Structure 83 for further reading on where to find the recorded data files.

3.2.2 Run Experiment

1. Fit the SMI-ETG to your participant's head (Hardware Setup 30).

2. Make sure your participant is wearing the SMI-ETG correctly. Please tighten the head strap to ensure a firm fit.

3. Launch iViewETG software.

4. Check whether the “Connected” symbol  appears in the upper right corner of the screen. This indicates that the SMI-ETG are properly

connected and ready to go. In case of other symbols appearing please see Hardware Setup 30.

5. Press “New Experiment…” to setup a new experiment (Creating a New Experiment 46), or press “Open Experiment…” to open an existing experiment (Opening an Existing Experiment 47)

6. Check if all relevant settings are according to your needs or change relevant settings (Experiment Run Settings 49)

7. Press “Record” to proceed to the Recording Screen 77.

8. Wait for the scene video to appear on the screen. This may take up to 30 seconds.

9. Press "New Respondent" to enter the name of the participant.



Before starting the calibration process, make sure that at least 30 seconds of time have passed since the participant has set up the SMI-ETG. This will assure the internal eye model has been completely adapted.

10. Perform a 1-point calibration:
    - 10.1 Press “1-Point-Calibration”.
    - 10.2 Ask your participant to look at a noticeable object in his field of view at a distance of 1.5m.
    - 10.3 Move mouse over scene video view. Move the crosshair to the object where the participant looks (or is looking) at and click on the object (1-Point Calibration 67).

    The system is now calibrated. You can check for calibration status in the Recording Display 78: The calibration status shows “1-Point Calibrated”.

11. Press "Record" to start the recording.

12. To stop the recording press "Record" again. When the red dot stops pulsating, the recording stops. The recorded data is getting saved automatically.

See Data Storage Structure 83 for further reading on where to find the recorded data files.

 (c) 2012 SensoMotoric Instruments

SMI Eyetracking Glasses Hardware

Chapter



4 SMI Eyetracking Glasses Hardware

This chapter will show how to set up the SMI-ETG system for mobile eye tracking studies.

4.1 Hardware Setup

1. Boot the SMI-ETG laptop. Login to Windows if necessary and make sure the battery is charged, ready and inserted properly.

2. Insert the non-shaded eye glasses into the SMI-ETG. In case of outdoor use with bright light, the shaded eye glasses can also be used (Convertible Eye Glasses 33). SMI does not recommend using the SMI-ETG without any of the eye glasses, since this can lead to a reduced tracking stability.

3. Adapt the eye position with the nose rest if necessary (Nose Rest 32).

4. Connect the USB plug of the SMI-ETG to one of the USB ports of the SMI-ETG laptop as shown below. The SMI-ETG will now initialize. This should take approximately 30 seconds. No video signal will be sent to the laptop during this phase.



Do not operate SMI-ETG ETG-Laptop without battery, even when connected to power supply. You may swap the battery when connected to power supply!



Do not connect any other USB device to the SMI-ETG laptop other than the BeGaze license dongle when SMI ETG is connected. Connecting additional devices may affect the performance of iViewETG.

 (c) 2012 SensoMotoric Instruments



Connection SMI-ETG ETG-Laptop via USB Port

5. Please put on the SMI-ETG and wear them as you would wear normal glasses. The temple arms will adjust to the head circumference.

Important: Do not overbend the temple arms! Do not fold the temple arms! Extending the temple arms beyond the mechanical stops will break the hinges! SMI will not hold responsible for damages to the frame which may affect the performance of the device!

6. Tighten the head strap as shown in the picture below. Optimal results will be accomplished by a steady fit of the SMI-ETG. Additionally the head strap will prevent the glasses from accidentally dropping.



Fixating the head strap

7. Make sure the participant feels comfortable when turning the head, nodding, looking around, walking etc.

8. Launch the iViewETG software by double clicking on the application icon.



Application Icon

9. The icons in the upper right corner of the screen indicate whether the SMI-ETG are working correctly. The following symbols may occur:

 No cameras found.

 No eye cameras found.

 No scene camera found.

 All cameras working.

 The software is not working correctly.

Now you are ready to prepare and perform a recording with SMI-ETG 13.

Be patient, allowing all sub systems to initialize correctly. This may take up to 30 seconds.(Troubleshooting 90)

4.1.1 Nose Rest

The SMI-ETG is designed to fit most participants when using the standard blue or red nose rest. Use nose rests in the following situations:

- Blue: For participants with narrower and more pronounced noses
- Red: For participants with wider and less pronounced noses

 (c) 2012 SensoMotoric Instruments

You can use the SMI-ETG without a nose rest as well.

Please change the nose rest in case your participant's eye is not positioned concentrically within the spectacle frame as this might affect tracking quality (Eye Position 73).



Two different nose rests. Insert one into the clamp and position carefully by moving up or down (snaps in).

Changing the nose rest is simple. Just carefully pull out the nose rest downwards from the nose bracket. Take the desired nose rest and carefully insert it into the clamp mounting. Make sure the nose rest snaps in, it will only work in the correct direction. After changing the nose rest, you can correct the eye position by slightly moving the nose rest up or down.

4.1.2 Convertible Eye Glasses

The convertible eye glasses reduce IR interferences caused by external light sources. SMI recommends always using one of the shaded or non-shaded glasses. Use the shaded eye glasses for brighter light conditions, e.g. during outdoor use.

When interchanging the eye glasses do not overbend them. For cleaning, please refer to Maintenance 7.



Non-shaded eye glasses. Inserting the shaded eye glasses

(c) 2012 SensoMotoric Instruments

iViewETG Software

Chapter



5 iViewETG Software

This chapter will show how to use iViewETG software for experimental design and recording.

There are two options to perform a recording using the iViewETG software:

- Quick Run 37: standard settings are used, data is stored separately for every participant
- Experiment Run 45: customized settings are used, data is stored in one experiment folder for all participants

See Data Storage Structure 83 for more information about storing the recorded data.

See Hardware Setup 30 for information on how to prepare the hardware before performing a Quick Run or an Experiment Run.

5.1 Starting iViewETG Software

Launch the iViewETG software by double-clicking on the application icon.



Application Icon

The dashboard opens which is your starting point for Quick Runs, Experiments and general settings of the iViewETG software. (User Interface 75).

(c) 2012 SensoMotoric Instruments



Dashboard

5.2 Preparing a Quick Run

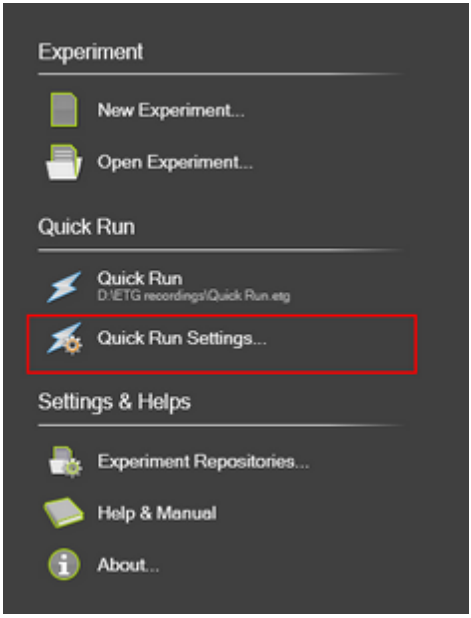
The easiest way to do first recordings is the "Quick Run" function of iViewETG. To adjust the settings of a Quick Run to your needs, you can change the predefined settings of a Quick Run.

5.2.1 Quick Run Settings

The iViewETG software is delivered with standard Quick Run settings which should provide optimal results of your recordings in most cases. Nonetheless, these settings can be adapted to your needs and for specific situations.

The "Quick Run Settings" will be used as standard settings for every Quick Run you perform. To adapt the settings for specific experiments, please refer to Experiment Run Settings 49.

To adapt the settings for Quick Runs press "Quick Run Settings" on the iViewETG dashboard:



Open Quick Run Settings from Dashboard

The Quick Run Settings window will appear which allows you to adjust Experiment Settings 53 and Hardware Configuration 55.



Quick Run Settings

Quick Run settings can also be imported from file. To load an existing setting file (*.etg) press "Import…". A file dialog will appear. Navigate to the directory where the according *.etg file resides. Select the file and press "Open". The settings stored in this file will then be loaded.

5.2.1.1 Default Calibration Mode

Select the calibration mode that you would like to be selected automatically when doing a Quick Run. You can choose between 1-Point Calibration and 3-Point Calibration. Please refer to Calibration 64 for further information about the different calibration modes.

5.2.1.1.1 0-Point Calibration

iViewETG will perform a 0-point calibration as soon as the eyes of the participant are being tracked and the Recording Screen 77 is entered. Therefore a gaze cursor will be shown automatically.

To increase accuracy you can perform a 1-Point Calibration 67 or 3-Point Calibration 68.



It is very important to provide a name for New Respondent 62 whenever a new participant starts wearing the SMI-ETG. This will adapt the internal eye model to the participant's eyes. Otherwise the gaze cursor may not stabilize and the results will be incorrect.

5.2.1.1.2 1-Point Calibration

1-point calibration is quick and easy to perform and should be your choice for fast and efficient data recording. Please refer to 1-Point Calibration 67 for further information on how to perform a 1-Point Calibration and when to use it.

5.2.1.1.3 3-Point Calibration

A 3-point calibration can result in more accurate data, but takes more time before the experiment task can be started. In case you are not satisfied with the result of a 1-Point Calibration you might want to try using a 3-Point Calibration to get more accurate results. Note: In most cases a 1-Point Calibration should provide the same accuracy as a 3-Point Calibration.

Please refer to 3-Point Calibration 68 for further information on how to perform a 3-Point Calibration and when to use it.

 (c) 2012 SensoMotoric Instruments

5.2.1.2 Allow Different Calibration Mode

Check this box to make calibration modes other than the default settings available during the Quick Run. If this option is not selected, only the Default Calibration Mode selected above will be available.

5.2.1.3 Eye Tracking Mode

SMI-ETG provides binocular 30Hz eye tracking. This configuration cannot be changed.

5.2.1.4 Scene Camera Mode

SMI-ETG provide HD scene camera mode (1280x960 pixel, 24fps). This configuration cannot be changed.

5.2.1.5 Scene Camera White Balance

The iViewETG software provides different white balancing modes. The Automatic White Balance mode adjusts white balance automatically, so that colors will match to what you see in real life. However, in some cases, especially under artificial (tungsten or fluorescent) light, the colors of objects might not match sufficiently. You can then select a different white balance mode according to the character of the light source.

5.2.1.5.1 Automatic White Balance

This is the default value of SMI settings and should provide optimal results in most cases.

5.2.1.5.2 Daylight (Outdoor)

Try this option if the color match in outdoor settings is distorted.

5.2.1.5.3 Tungsten

A white balancing with tungsten means a white balancing in a normally "warm" light, mostly found indoors with incandescent bulb lamps. These bulbs give a yellowish-orange light..

5.2.1.5.4 Fluorescent

Fluorescent light is perceived as more "cold" and you will find this kind of light more frequently in offices, laboratories, or public places with a character of work areas. Mostly, the light is generated by neon lamps.

5.2.1.5.5 Fluorescent High

Use this setting in very bright fluorescent environments.

5.2.1.6 Microphone

Check this box to add audio recording using the integrated microphone of the SMI-ETG.

Currently the microphone is activated by default and cannot be switched off.

5.2.2 Starting a Quick Run

To start a Quick Run please follow these steps:

1. Fit the SMI-ETG to the participant's head (Hardware Setup 30).
2. Make sure the participant is wearing the SMI-ETG correctly. Please tighten the head strap to ensure a firm fit.
3. Launch iViewETG software.

 (c) 2012 SensoMotoric Instruments

4. Check whether the “Connected” symbol  appears in the upper right corner of the screen. This indicates that the SMI-ETG are properly connected and ready to go. In case of other symbols appearing please see Hardware Setup 30.

5. Press "Quick Run" on the iViewETG dashboard. The Recording Screen 77 will appear.

6. Wait for the scene video to appear on the screen. This may take up to 30 seconds.

7. Press "New Respondent" to enter the name of the participant.

It is very important to provide a name for “New Respondent” whenever a new participant starts wearing the SMI-ETG. This will adapt the internal eye model to the participant's eyes. Otherwise the gaze cursor may not stabilize and the results will be incorrect.

5.2.3 Calibration

Before starting the calibration process, make sure that at least 30 seconds of time have passed since the participant has set up the SMI-ETG. This will assure the internal eye model is completely adapted.

Press the button that is either labeled “1-Point Calibration” or “3-Point Calibration” to start the calibration process. Ask the participant to look at a noticeable object in his field of view at a distance of 1.5m. Move the mouse over the scene video view. Move the crosshair to the object where the participant is looking at and click on the object. In case of a 3-Point Calibration proceed in the same way for the rest of the calibration points.

The system is now calibrated. You can check for calibration status in the

Recording Display 78: The calibration status shows “1-Point Calibrated” or “3-Point Calibrated”.

5.2.4 Quick Run Recording

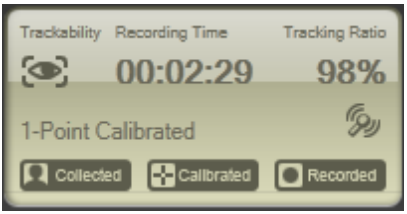
After fitting the glasses to your participant, typing in the name and performing a calibration, you can start your first Quick Run recording.

Click on 

on the Recording Screen 77 to start data recording.

During the recording the red light of the recording button will pulsate smoothly. To stop the recording press "Record" again. When the red dot stops pulsating, the recording stops. The recorded data is getting saved automatically.

The Recording Display 78 provides information about the recording process:



Recording Display

You can now proceed in two different ways:

Perform additional recording with the same participant
To perform another recording with the same participant, simply press "Record" again. The additional recording will be stored in the same folder.

Perform additional recording with a new participant
In order to perform a recording with a new participant select the

green plus-symbol next to the current participant's name. Typing in a new participant's name will reset the current calibration!



Provide name of a new participant

Please perform a calibration when you have selected a new participant. Press “Record” to start the recording again.

To stop the recording press "Record" again. When the red dot stops pulsating, the recording stops. The recorded data is getting saved automatically.

See Data Storage Structure 83 for further reading on where to find the recorded data files.

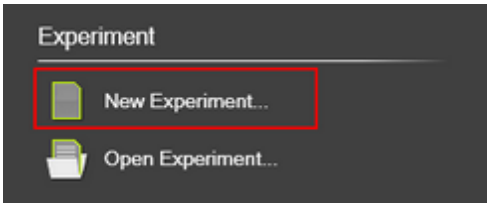
5.3 Preparing Experiment Run

To prepare an experiment, it is necessary to know beforehand which properties you intend to collect in order to perform your analysis. The requirements for specific study designs can be defined in your Experiment Settings 53. You may change settings between recording sessions. Please keep in mind that this might have an impact on the results and the analysis of the recorded data.

5.3.1 Handle Experiments

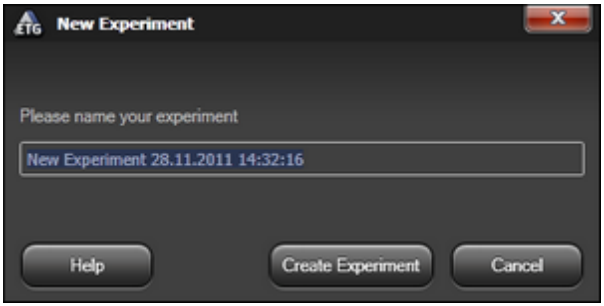
5.3.1.1 Creating a New Experiment

Click "New Experiment" on the dashboard.



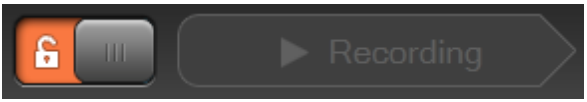
Create new Experiment

You will be prompted to enter a name for the new experiment. Click "Create Experiment" when you have selected a proper experiment name.



Name Experiment

The Experiment Settings screen will open. Please refer to Experiment Run Settings 49 for further information on how to add subject properties, change experiment settings or Hardware Configuration 55. Changes to the settings will be saved to the experiment automatically. Lock the experiment by clicking on the symbol in the upper right corner of the screen. Settings can be locked to prevent unintentional changes during experiment phases.



Settings unlocked - you can change settings but no recording is possible



Settings locked - recording can be started

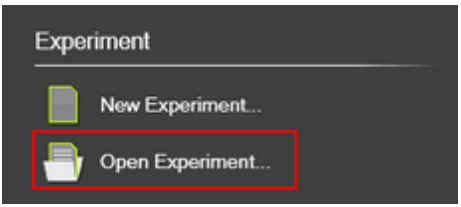
When the experiment is "locked" you can proceed with the experiment by clicking on the "Record" button. The recording screen will appear.

Please be aware that initializing the SMI-ETG can take up to 30 seconds. Be patient, allowing all sub systems enough time to initialize and to respond to one another appropriately.
(Troubleshooting 90)

To continue with the current experiment please go to Starting an Experiment Run 57.

5.3.1.2 Opening an Existing Experiment

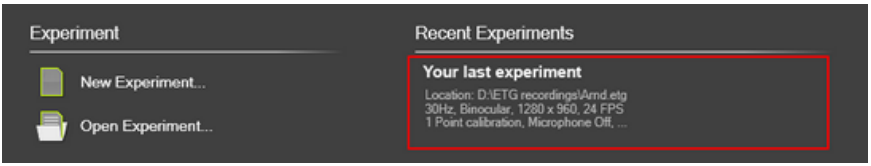
You can open an existing experiment by clicking on "Open Experiment" on the iViewETG dashboard.



A file dialog will open and will ask you to select an iViewETG experiment

file. Navigate to the directory where the according *.etg file resides. Select the file and press “Open”. The experiment will then be loaded.

The easiest way to open recently used experiments is to select the experiment from the "Recent Experiments" section on the dashboard. Click on the desired experiment description to load the related experiment.



The Settings Screen 76 will appear. You can change settings now if you wish to adapt the loaded experiment to your needs.

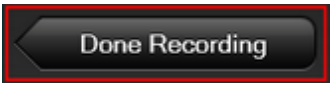
Lock the experiment when you are finished with adapting the settings. You can now press “Record”



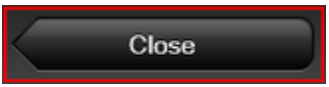
to proceed to the Recording Screen 77. Please refer to Starting an Experiment Run 57 for further information on how to start the recording.

5.3.1.3 Close Experiment

To close an experiment you have to stop any currently running recordings. The recorded data is getting saved automatically. Press “Done Recording” to close the experiment.



The Settings Screen 76 will appear. Here you can have a final overview of the settings being used for recording. Press “Close” to return to the iViewETG dashboard.



You can now either conduct another study or close the software by pressing "Quit iViewETG ".

See Data Storage Structure 83 for further reading on where to find the recorded data files. Please refer to the SMI BeGaze manual for detailed information on how to analyze the recorded data.

5.3.2 Experiment Run Settings

The iViewETG software is delivered with standard experiment settings which should provide optimal results of your recordings in most cases. Nonetheless, these settings can be adapted to your needs and to specific situations.

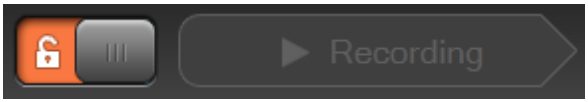
To adapt the settings for a specific experiment, load the according experiment or create a new one (Creating a New Experiment 46). Unlock the experiment in case it should be locked. Press “Change” in the desired section to change the settings.

5.3.2.1 Lock/Unlock Settings

Before you can change settings you need to unlock the settings. You find the corresponding switch in the upper right corner of your settings screen.



The settings are locked - you are able to start the recording



The settings are unlocked - you can change settings but not record

5.3.2.2 Participant Properties

It is possible to add participant properties for each participant. The property settings data will be stored to your experiment file and will be available for this experiment only. The data of the participant's properties itself will be saved to the participant's data file to be used for the analysis.

Enter participant properties either when typing in the participant's name or at any time simply by clicking on the button with the participant's name. Both will open the dialog to enter properties. You can also add or change properties in the SMI BeGaze Analysis Software. Deleted properties will still be available in already recorded datasets.

All properties you set are optional, and is not possible to mark fields as required properties. So be aware your team knows about this and enters the data.

Adding a new participant in the recording screen will look similar (depending on your experiment participant properties) to the following picture:



participant properties in recording screen

5.3.2.2.1 Add Properties

To add properties unlock the settings, click "Change" next to the participant properties and you will be prompted with the "participant Property Definition", which allows you to add, edit or delete the participant properties. If you haven't defined a property by now, the software will ask you to click "Add" in order to add a first data row.



Properties will be listed under "Property Name" and "Value Candidates"

To define a new property simply click "Add" and a new line will appear. Position the cursor into the fields and type the name and the values of the desired properties. Basically, the properties are stored as key (Property Name) and value (Value Candidates) sets whereby one key may be related to one value. In order to keep this process easy and efficient later on during the experiment, you can select the value from a drop down list. E.g. it is possible to add a property name "gender" and set the selectable values to "male" or "female". Just type in the desired values and separate them with a semicolon (;). During data collection, when adding a participant for recording you can now select from "male" or "female" at the property "gender". (Technical Specifications 19)

Alternatively, if there are no Value Candidates defined, you may enter free text as individual property value for each participant.

Click "OK" and the property is set.

In order to add more properties to existing ones, just click the "Add" button and a new line with empty fields will appear.

5.3.2.2.2 Edit Properties

Editing and changing existing properties is very simple. Unlock the settings, click "Change" on the participant Settings, position the cursor in the properties name you wish to edit and just type, change or add the desired information at the property name or Value Candidates' column. Click "OK" and the property is edited..

5.3.2.2.3 Delete Properties

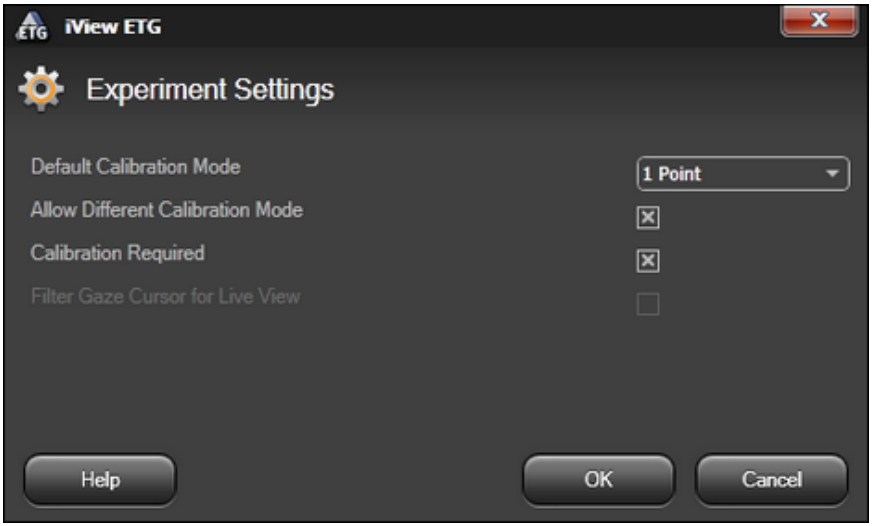
Deleting properties is as simple as adding or editing them. Just open the "participant Properties Window", click the line you wish to delete and click "Delete".



After clicking "OK" the property is deleted. You will not be asked whether you really wish to delete the data!

5.3.2.3 Experiment Settings

In order to change the experiment settings, unlock the settings and click "Change..." right to the "Experiment Settings". You will be prompted with the "Experiment Settings" window.



Experiment Settings

5.3.2.3.1 Default Calibration Mode

Select the calibration mode that you would like to be selected automatically when running an experiment. You can choose between 1-Point Calibration and 3-Point Calibration. Please refer to -> Calibration for further information about the different calibration modes.

5.3.2.3.1.1 0-Point Calibration

iViewETG will perform a 0-point calibration 65 as soon as the eyes of the participant are being tracked and the Recording Screen 77 is entered. Therefore a gaze cursor will be shown automatically.

To increase accuracy you can perform a 1-Point Calibration 67 or 3-Point Calibration 68.

 (c) 2012 SensoMotoric Instruments



It is very important to provide a name for New Respondent 62 whenever a new participant starts wearing the SMI-ETG. This will adapt the internal eye model to the participant's eyes. Otherwise the gaze cursor may not stabilize and the results will be incorrect.

5.3.2.3.1.2 1-Point Calibration

A 1-point calibration is quick and easy to perform and should be your choice for fast and efficient data recording. Please refer to 1-Point Calibration 67 for further information on how to perform a 1-Point Calibration and when to use it.

5.3.2.3.1.3 3-Point Calibration

A 3-point calibration can result in more accurate data, but takes more time before the experiment task can be started. In case you are not satisfied with the result of a 1-Point Calibration you might want to try using a 3-Point Calibration to get more accurate results. Note: In most cases a 1-Point Calibration should provide the same accuracy as a 3-Point Calibration.

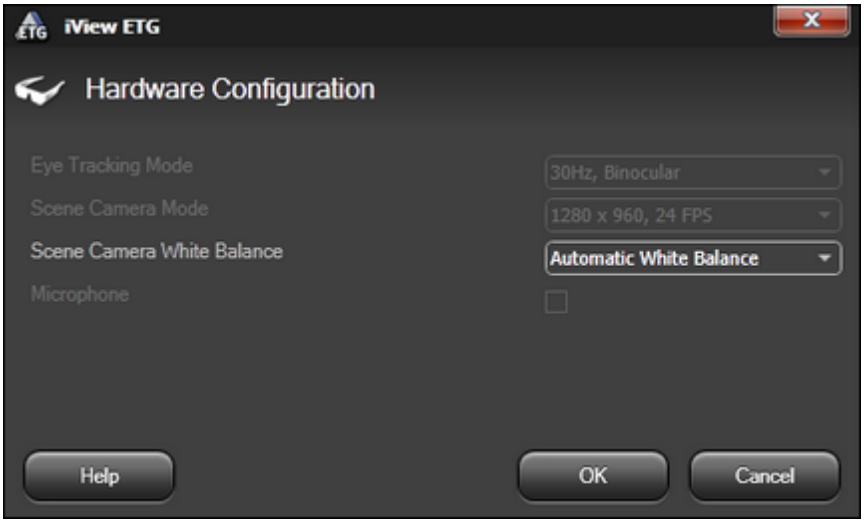
Please refer to 3-Point Calibration 68 for further information on how to perform a 3-Point Calibration and when to use it.

5.3.2.3.2 Allow Different Calibration Mode

Check this box to make calibration modes other than the default one available during the Quick. If this option is not selected, only the Default Calibration Mode selected above will be available.

5.3.2.4 Hardware Configuration

SMI-ETG provide a binocular 30Hz eye tracking with HD scene camera mode (1280x960 pixel, 24fps). This configuration cannot be changed.



Hardware Configuration

5.3.2.4.1 Scene Camera White Balance

The iViewETG software provides different white balancing modes. Normally, SMI-ETG automatic white balance corrects the balance and the colors match to what you see in real life. However, in some cases, especially under artificial (tungsten or fluorescent) light, the colors of objects might not match. If you experience this phenomenon you can correct the white balance according to the character of the light source.

5.3.2.4.1.1 Automatic White Balance

This is the default value of SMI settings and should provide optimal results in most cases.

5.3.2.4.1.2 Daylight (Outdoor)

Try this option if the color match outdoors is distorted.

 (c) 2012 SensoMotoric Instruments

5.3.2.4.1.3 Tungsten

A white balancing with tungsten means a white balancing in a normally "warm" light, mostly found indoors with incandescent bulb lamps. These bulbs give a yellowish-orange light.

5.3.2.4.1.4 Fluorescent

Fluorescent light is perceived as more "cold" and you will find this kind of light more frequently in offices, laboratories, or public places with a character of work areas. Mostly the light is generated by neon lamps.

5.3.2.4.1.5 Fluorescent High

Use this setting in real bright fluorescent environments.

5.3.2.4.2 Microphone

Check this box to add audio recording using the integrated microphone of the SMI-ETG.

Currently the microphone is activated by default and cannot be switched off.

5.3.3 Starting an Experiment Run

This chapter provides some information about the process of running an experiment. At first, you create 46 or open 47 an experiment. If necessary change the settings adapting hardware configuration 55, experiment settings 53, and participant properties 50 to the values which fit best to the requirements of your experiment.

Then lock the settings 49 and you're ready to start with the recording 44.

1. Fit the SMI-ETG to your participant's head (Hardware Setup 30).

2. Make sure your participant is wearing the SMI-ETG correctly. Please tighten the head strap to ensure a firm fit.

3. Launch iViewETG software.

4. Check whether the “Connected” symbol  appears in the upper right corner of the screen. This indicates that the SMI-ETG are properly connected and ready to go. In case of other symbols appearing please see Hardware Setup 30.

5. Press “New Experiment…” to setup a new experiment (Creating New Experiment 46), or press “Open Experiment…” to open an existing experiment (Opening an Existing Experiment 47).

6. Check if all relevant settings are according to your needs or change relevant settings (Experiment Settings 53).

7. Press "Record" to proceed to the Recording Screen 77.

8. Wait for the scene video to appear on the laptop screen. This may take up to 30 seconds.

9. Press "New Respondent" to enter the name of the participant.



Please be aware that initializing the SMI-ETG can take up to 30 seconds. Be patient, allowing all sub systems enough time to initialize and to respond to one another appropriately. (**Troubleshooting** 90)

Check whether this symbol appears in the upper right corner of the screen.

 This indicates that the SMI-ETG are properly connected and ready to go. Make sure your participant is wearing SMI-ETG correctly. Please read the section "Hardware Setup 30" to learn how to prepare the hardware before running an experiment.

5.3.4 Calibration

Before starting the calibration process, make sure that at least 30 seconds of time have elapsed since the participant has set up the SMI-ETG. This will assure the internal eye model is completely adapted.

Press the button that is either labeled "1-Point Calibration" or "3-Point Calibration" to start the calibration process. Ask the participant to look at a noticeable object in his field of view at a distance of 1.5m. Move the mouse over the scene video view. Move the crosshair to the object where the participant looks at and click on the object. In case of a 3-Point Calibration proceed in the same way for the rest of the calibration points.

The system is now calibrated. You can check for calibration status in the Recording Display 78: The calibration status shows "1-Point Calibrated" or "3-Point Calibrated".

5.3.5 Experiment Run Recording

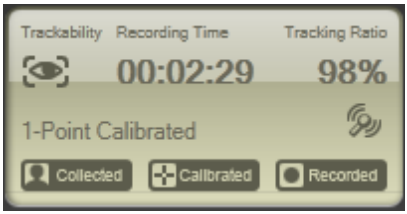
After fitting the glasses to your participant, typing in the name and performing a calibration, you can start your first Experiment Run recording.

Click on 

on the recording screen to start data recording.

During the recording, the red light of the recording button will pulsate smoothly. To stop the recording press "Record" again. When the red dot stops pulsating, the recording stops. The recorded data is getting saved automatically.

The Recording Display provides information about the recording process:



Recording Display

You can now proceed in two different ways:

Perform additional recording with the same participant
To perform another recording with the same participant, simply press "Record" again. The additional recording will be stored in the same folder.

Perform additional recording with a new participant
In order to perform a recording with a new participant select the green plus-symbol next to the current participant's name. Typing in a new participant's name will reset the current calibration!



Provide name of a new participant

Please perform a calibration when you have selected a new participant.
Press “Record” to start the recording again.
To stop the recording, press "Record" again. When the red dot stops pulsating, the recording stops. The recorded data is getting saved automatically.

See Data Storage Structure 83 for further reading on where to find the recorded data files.

 (c) 2012 SensoMotoric Instruments

In-Depth Information

Chapter



6 In-Depth Information

6.1 New Respondent

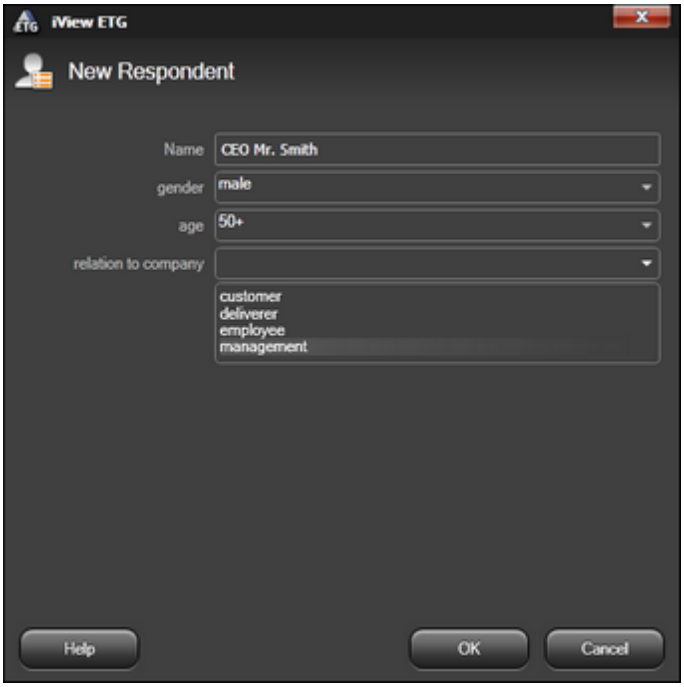
Click "New Respondent", or alternatively the green cross if you have previously entered a participant



Starting recording for a new participant

Fill in the name and participant properties. To learn how to define participant properties, please read participant Properties 50.

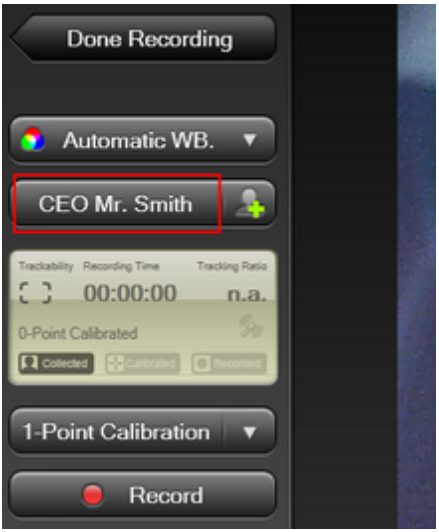
 (c) 2012 SensoMotoric Instruments



New participant's name and properties

Click "OK" to continue.

You will now see of your participant's name on the recording screen.



Name of participant on the video screen

Selecting a new respondent will reset the internal eye model to 0-point calibration. Afterwards the internal eye model will be adapted to the new participant'seyes. This will assure optimal tracking stability and accuracy.

6.2 Calibration

Different calibration options are available with iViewETG software.

- 0-point calibration 65
- 1-point calibration 67
- 3-point calibration 68

You may also perform an off-line calibration using SMI BeGaze Analysis Software. Please read the SMI BeGaze™ manual which is included in the delivery of SMI-ETG for more information.

In case the warning "insufficient trackability of the subject" appears, please refer to the section Gaze Cursor Stabilization 72.

Type in the name of a new participant first and then perform a 1-point or 3-point calibration.

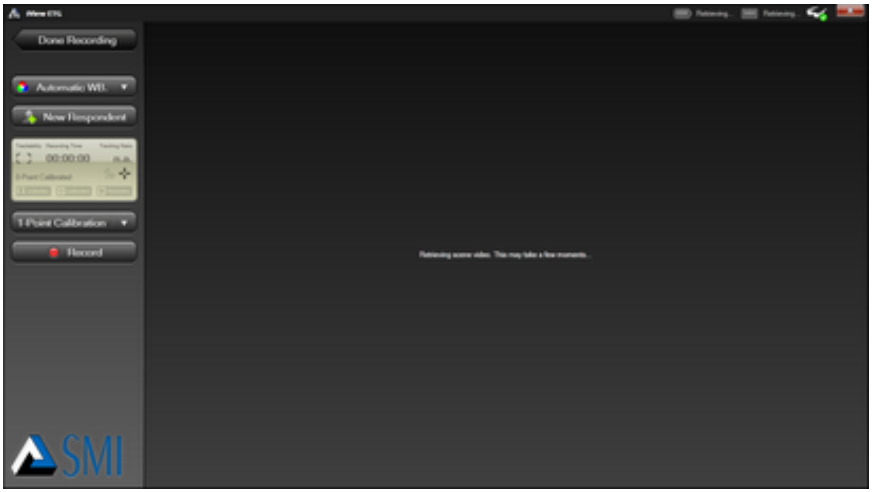
Typing in the name of a new participant resets the calibration!

Before starting the calibration process, make sure that at least 30 seconds of time have passed since the participant has set up the SMI-ETG. This will assure the internal eye model is completely adapted.

6.2.1 0-Point Calibration

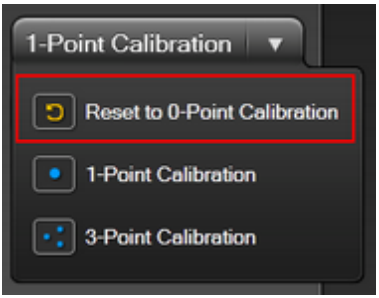
SMI-ETG and iViewETG software will conduct the 0-point calibration based on the SMI eye model after entering the recording screen. It may take up to 30 seconds before the model has fully adapted to the participant's eye physiognomy. Make sure the participant is moving the eyes around during this calibration phase to ensure better adaptation.

In case you do not see a gaze cursor or/and the warning "insufficient trackability of the subject" appears, please check for gaze cursor stabilization (Gaze Cursor Stabilization 72).



Retrieving thescene video signal

Whenever a new participant starts wearing the SMI-ETG the internal eye model will have to be adapted to the participant's eyes. This happens automatically when a name for a New Respondent 62 is given. You can also force the adaptation to be started by selecting "Reset to 0-Point Calibration".



Once the 0-point calibration delivers optimal or sufficient results of gaze stability and accuracy, your SMI-ETG are ready to go and you may

continue and start your first recording session . Should the 0-point calibration not provide sufficient results and the gaze cursor not match theparticipant's real gaze position, please proceed with 1- or 3-point calibration (1-Point Calibration 67 or 3-Point Calibration 68).



Blue gaze cursor

6.2.2 1-Point Calibration

Click on 

and the 1-point calibration will start immediately. Move the cursor over the scene camera image and you see crosshairs in addition to the gaze cursor. The numbers beside the crosshairs indicate how many points for the calibration are necessary (right number) and how many points you still need to click (left number). In case of the one-point-calibration this number will be 1/1.

Ask your participant to look at a noticeable object in his field of view at a distance between the participant and the calibration object of 1.5m.

Your participant should sit or stand comfortably while looking straight

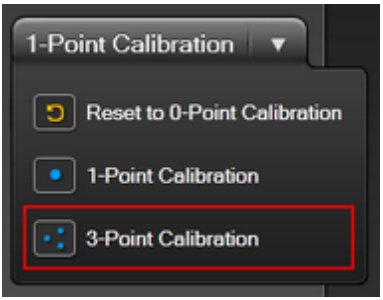
ahead without moving the head. The object should be at the middle of the scene camera image for optimal accuracy. Click with the crosshairs on this object on the laptop screen. The gaze cursor's offset will now be corrected to the crosshair's position.

Validate the accuracy of this calibration by asking your participant to look on different objects from their present location. Ideally, choose some objects nearby and at the distance you want to measure. If the gaze cursor matches the objects, you can continue by starting the recording session 44. Otherwise, you may retry the 1-point calibration or run a 3-point calibration 68.

To abort a calibration, just click the highlighted calibration button again.

6.2.3 3-Point Calibration

Start the calibration process by clicking on the small arrow and select "3-Point Calibration". If "3-Point Calibration" is already displayed at the button just click it once and the 3-point calibration will start immediately.



3-Point Calibration

Move the cursor over the scene camera image and you see crosshairs in addition to the gaze cursor. The numbers beside the crosshairs indicate how many points for the calibration are necessary (right number) and how many points you still need to click (left number). In case of the first point of a 3-point-calibration this number will be 1/3.

 (c) 2012 SensoMotoric Instruments

Ask your participant to look at the first noticeable object in his field of view at a distance of 1.5 m between the participant and the calibration objects.

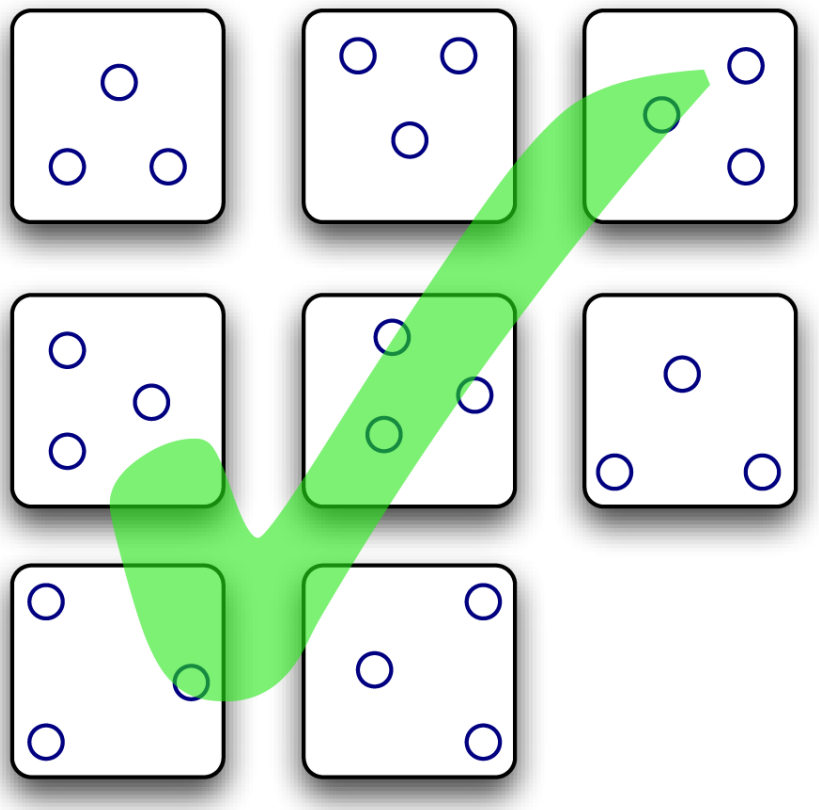
Your participant should sit or stand comfortably while looking straight ahead without moving the head. The objects should be spread over the scene camera image for optimal accuracy. Click with the crosshairs on the objects on the laptop screen.

In principal, a 3-point calibration should use a triangular position of stimulus points, which are on the same plane. Do not try to calibrate by using 3 points in a row and / or at different distances from the participants eye. To perform the calibration just start with clicking on any of your calibration points the participant is looking at. It is not required to follow a dedicated order in calibration.
You are now ready to start your recording session 44.

To abort a calibration, just click the highlighted calibration button again.

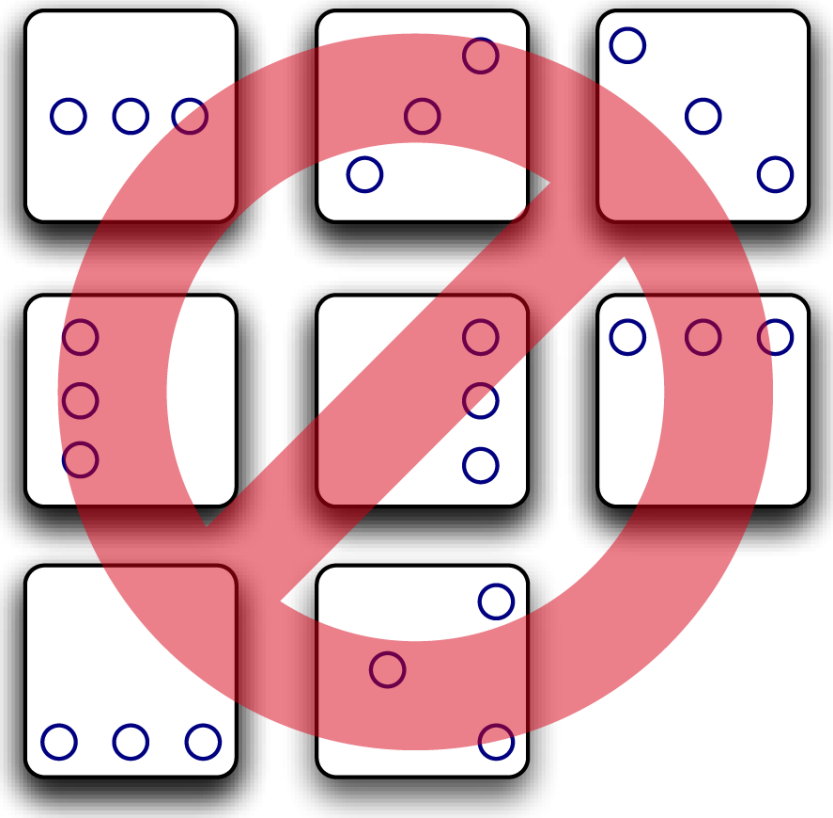


It is very important to provide a name for New Respondent whenever a new participant starts wearing the SMI-ETG. This will adapt the internal eye model to the participant's eyes. Otherwise the gaze cursor may not stabilize and the results will be incorrect.



Please calibrate setting the stimulus points like shown in the figure above

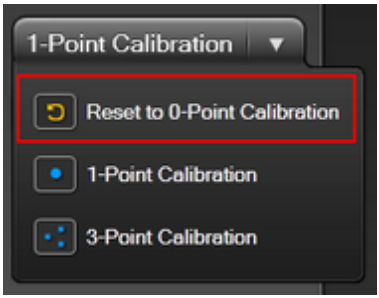
(c) 2012 SensoMotoric Instruments



Calibration in these configurations will not work properly

6.2.4 Calibration Reset

It is very important to provide a name for New Respondent whenever a new participant starts wearing the SMI-ETG. This will adapt the internal eye model to the participant's eyes. Otherwise the gaze cursor may not stabilize and the results will be incorrect.



Select reset to 0-Point Calibration

6.3 Gaze Cursor Stabilization

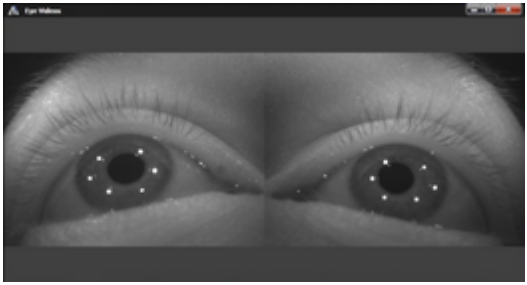
In some cases when trying to focus on a stimulus the cursor bounces around the screen or is stuck to the upper left corner of the screen or/and a warning "insufficient trackability of the subject" may appear. This indicates that the general process of getting sufficient information about the participant's eye movement is disturbed because of one of the following reasons.

1. 0-point calibration 65 has not been reset or failed to reset
2. Eye positioning 73 in the frame of the glasses is not correct
3. Monocular vision 73
4. Distortion caused by eye make-up 73 (mascara/eyeliner?)
5. Extreme ocular dominance 74 of one eye
6. Distracting IR reflections 74
7. Correction glasses 74 between the SMI EyeTracking Glasses and the eyes
8. Contact lenses 74

You need a stable gaze cursor to continue.

6.3.1 Eye Position

SMI-ETG are designed to fit most participants when using the standardized nose rests. To control the eye position press "Alt+Ctrl+E" after entering the recording screen 77 and the two eye images will show you the position of the eye as recognized by SMI-ETG. You can adapt the eye position by changing the nose rest 32.



Eye Image

6.3.2 Monocular Vision

SMI-ETG are a binocular mobile eye tracking device. The system tracks both eyes of a participant. In cases of monocular vision, e.g. due to an artificial eye, the system will not receive enough information to calculate the gaze position and the system will not acquire sufficient tracking information for that particular participant.

6.3.3 Make Up

SMI-ETG are using infrared light during the eye tracking process. In some cases eye makeup (especially mascara) may disturb correct measurements. Please ask your participant to remove make up..

6.3.4 Ocular Dominance

In binocular vision, the dominant eye is the one which is primarily relied on for positional information. If the dominant eye is 'overdominant', this normal effect can sometimes interfere with the SMI Eye Tracking Glasses' measurement model. In these cases, the system will not acquire sufficient tracking information for that particular participant.

6.3.5 Distracting IR Reflection

Always use the convertible eye glasses delivered with the SMI-ETG (Eye Glasses 33) to avoid distracting IR reflection from external sources in your participant's eyes, which may lead to inaccurate eye tracking or loss of trackability. Also read the instructions on the use of corrective glasses 74 and contact lenses 74.

n extremely bright light conditions it may be necessary to block IR light from above by wearing a cap or something similar above the glasses.

6.3.6 Corrective Glasses

SMI-ETG hardware are not designed for use with normal corrective glasses. The inclusion of participants wearing corrective glasses needs to be verified on a case by case basis.

6.3.7 Contact Lenses

In most cases, soft and hard contact lenses work perfectly well with the SMI-ETG. In some cases, contact lenses may disturb correct tracking results. Should you encounter problems, if possible, ask your participant to remove them.

 (c) 2012 SensoMotoric Instruments

6.4 User Interface

The user interface of iViewETG software has been kept as simple as possible to ensure most efficient performance of experimental and recording field situations.

6.4.1 Dashboard

The dashboard is your starting point for Quick Runs, Experiments and general settings of the iViewETG software. It is intuitive and you will find four main topics:



1. Experiments 45
2. Recent Experiments 47
3. Quick Run 42
4. Settings 53 & Help

5. Quitting iViewETG

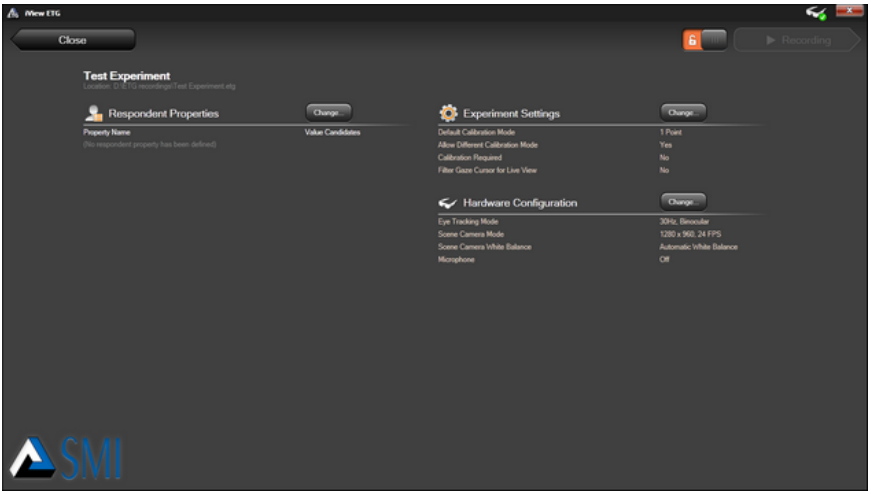
Please see the correspondent topics to learn more about it.

6.4.2 Settings Screen

The settings screen displays the current settings chosen for your Experiment. It contains the Participant Property Settings 50, the Experiment settings 53, and the Hardware Configuration 55.

To change settings, you need to unlock the setting screen with the lock icon in upper right corner. The "Change..." buttons will get active. Lock it again to continue to the recording screen. If you press the close button in the upper left corner, your experiment is saved and closed. (See Experiment Settings 53)

On the left side you will find the name of the experiment. It is located above the storage location on your disk.



Unlocked settings screen

6.4.3 Recording Screen

You will enter the recording screen when performing a quick run or running an experiment. You will see the name of the experiment or a quick run displayed in the upper left corner of the screen.



Recording Screen

On the left you will find from top to bottom:

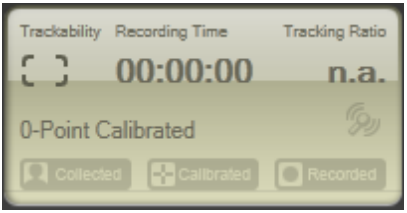
1. The closing button "Done Recording" - Click to close the recording screen and continue with the settings screen
2. White Balance Button - Click to change the white balance mode via drop down list (Scene Camera White Balance 41)
3. New participants Button - Click the green plus symbol to add a new participant. If you already added one, the button provides you with the participant's name (New participant 62)
4. The Recording Display (Recording Display 78)

5. The Calibration Button (Calibration 64)
6. The recording button - Click once to start the recording, click again to stop the recording.

On the top bar you find information about the battery life, the remaining storage, and the status of SMI-ETG (Runtime Information 83).

6.4.4 Recording Display

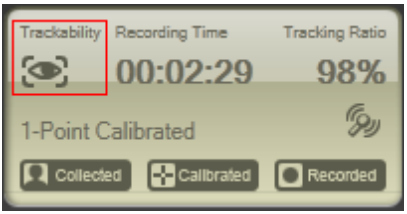
The main visual display on the recording screen is the recording display. It assembles all information about the ongoing recording process; including the trackability of the participant with SMI-ETG, recording time, tracking ratio, calibration type, microphone setting and three steps of confirmation during the individual participant'srecording process.



Light gray icons indicate that this feature is not available or not in use at the moment

6.4.4.1 Trackability

The trackability indicates whether SMI-ETG are able to track the participant's eye movements correctly. This is not an indication of whether the calibration works correctly.



participant's eyes are tracked successfully

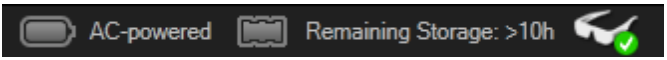
6.4.4.2 Recording Time

Generally the maximum recording time is defined by availability of disk space and battery life. SMI recommends a maximum of two hours recording time for one study on battery power.

Before starting your recording, please make sure that there is enough HDD space and battery life left for your estimated recording time. iViewETG software estimates the time of both and then displays its information in the upper right corner of the recording screen. A warning message will appear in case the software estimates less than 10 minutes battery or storage capacity.

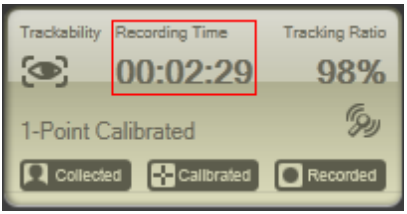


Battery Life (fully charged)



AC-powered and a remaining storage for more than 10h of recording time

The length of the individual recording can be checked on the recording display showing the recording time.

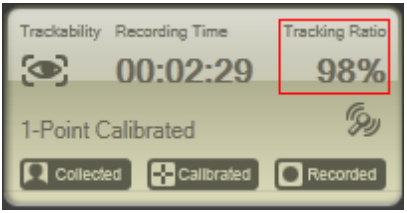


participant's recording time of 26 seconds.

6.4.4.3 Tracking Ratio

The tracking ratio is the ratio between total recording time and the time with valid tracking of the eyes. It indicates how well your participant's gaze could be tracked during the experiment. You will not get a tracking ratio of 100% during longer recordings, due to e.g. the participant blinking or closing the eyes. Also other influences may cause weaker trackability of your participant's eye movement.

During an experiment, the recording screen displays the tracking ratio valid for a time slot of 10 seconds prior to the current time. After the recording stops, the iViewETG software calculates the average tracking ratio of the complete session and displays the ratio.

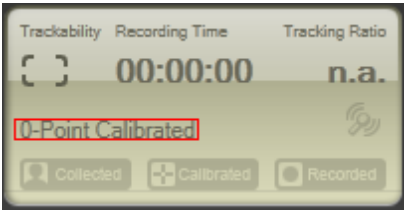


Tracking Ratio

6.4.4.4 Calibration

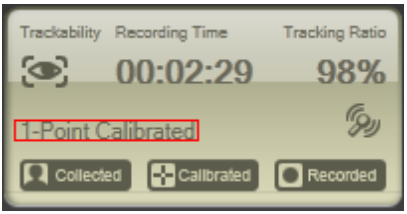
The calibration indication shows, which mode of calibration 64 has been performed. If you haven't conducted any calibration and trackability is available, the system uses a 0-point calibration, which is performed by SMI-ETG initially. If you pass the glasses to the next participant you need

to manually reset the 0-point calibration or to type in a new participant's name to automatically reset the software in order to get correct results and measurements.



Initial 0-Point Calibration

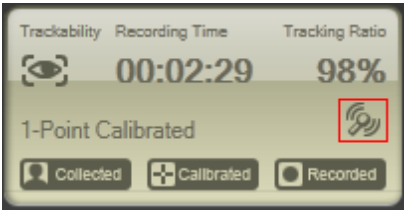
After the calibration is finished the indication shows which mode of calibration has been used.



1-Point Calibration after recording

6.4.4.5 Microphone

The recording display shows whether the microphone is switched on or off. (Microphone 57).



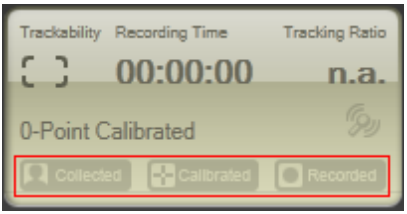
Microphone on

6.4.4.6 3-Step-Confirmation

iViewETG software supports you during your data collection by indicating the status of the three most important steps.

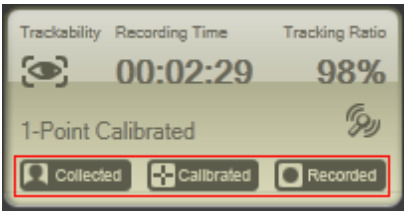
- Have you already collected the participant's properties 50?
- Have you performed a 1-Point 67 or 3-Point Calibration 68 (not taking into account 0-Point Calibration 65, which is done initially)
- Have you already recorded data?

The recording display starts with three light gray icons, indicating that nothing has been performed and set yet.



No recording, calibration and data collection has been performed

Whenever you step through your process, add a new participant with its properties, calibrate, or start and stop recording, the equivalent icon will turn to dark gray indicating, that this task has been accomplished.



In this case, everything has been done and accomplished - ready!

(c) 2012 SensoMotoric Instruments

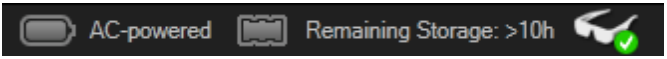
6.4.5 Runtime Information

In addition to the recording display which reflects the information you need, the following important runtime information is shown in the upper right corner of the recording screen.

- Battery or power information
- Remaining recording time
- Status of connection of SMI-ETG



Battery powerd



AC-powered

Please be aware that iViewETG only check HDD space available on drive D:\!

6.5 Data Storage Structure

There are two options to perform a recording using the iViewETG Software:

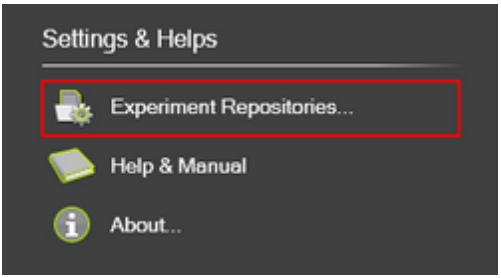
- Quick Run 37: A quick run uses standard settings and stores single user raw data (Directory Structure 86)
- Run Experiment 26: An experiment run requires dedicated settings and stores data of all recorded participants in one experiment folder (Directory Structure 86).

The default experiment repository path is set to "D:\SMI-ETG-Recordings\Experiments" and default quick run path is "D:\SMI-ETG-Recordings"



6.5.1 Change Data Storage Directory

In order to change your local data storage directory click on "Experiment Repositories" on the dashboard.

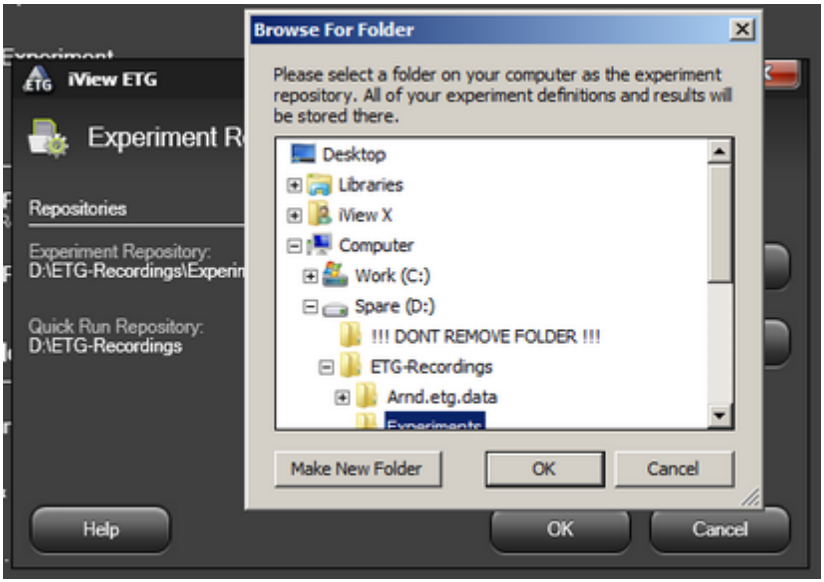


You will be prompted with the experiment repositories window.

 (c) 2012 SensoMotoric Instruments



Click "Change" and a system directory navigation window will open.

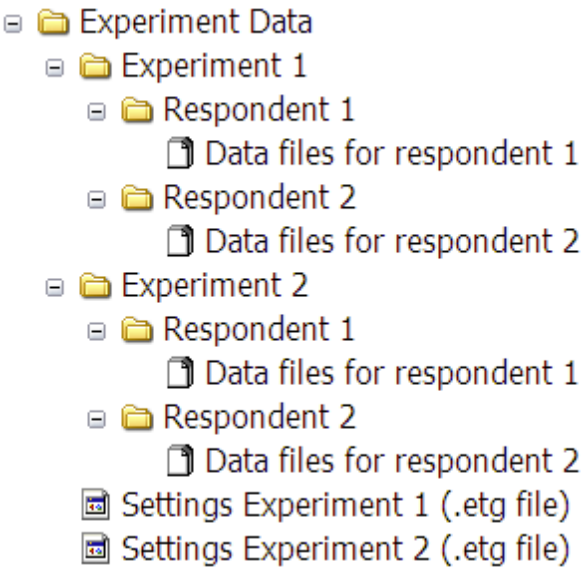


Navigate to the directory or subdirectory in which you want to store your

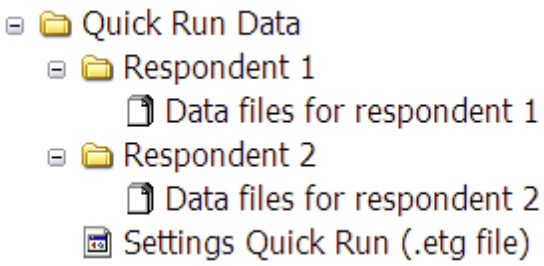
experiments, select a directory or create a new one and select it afterwards. Click OK and your experiments and the experiment data will be stored at this place.

6.5.2 Directory Structure

iViewETG software collects all your data of experiments and quick runs in your chosen directory or its initial directory, which is set by SMI on delivery. The general structure for experiments and quick run appears like shown in the following figures.



Experiment Directory Structure



Quick Run Directory Structure

participant directories' names within the experiment directories are concatenated with the participant's name and a unique directory name hash. Therefore, if you enter participants with identical names, no naming conflict will occur.

The experiment files (*.etg) contain all necessary information about the experiment. Do not delete or remove any of these files!

6.5.3 IDF Files

IDF (intermediate data file) files are a binary file format which stores information about the participant's complete raw gaze data. This data can only be converted to an ASCI text file with SMI BeGaze Analysis Software.

6.5.4 Video Files

Your experiment contains three video files and one audio file. The audio file will be saved as a .wav file. Two video files (.avi) are down sampled eye videos of the right and left eye only for support reasons.

6.6 Software Updates

Frequently check on the SMI support webpage to see if a new version of iViewETG or BeGaze software is available for download. If yes, please download the software to a virus protected storage device. Transfer it to the SMI-ETG laptop by using this device. Make sure that the involved devices are virus free to not risk a contamination of the SMI-ETG laptop.

Execute the installer on the SMI-ETG laptop and follow the further instructions on the screen.

Do not connect SMI-ETG laptop to the internet. No antivirus software is, nor should be, installed. Installing anti-virus software might harm measurement performance and measurement quality of the SMI-ETG system.

The SMI-ETG accompanying software can be downloaded from the following SMI website: http://www.smivision.com/software.

For license information see License Agreement and Warranty 116

 (c) 2012 SensoMotoric Instruments

Appendix

Chapter



7 Appendix

7.1 Troubleshooting

- Please be aware that the SMI-ETG needs time to boot. Be patient, allowing all sub systems to initialize correctly. This may take up to 30 seconds. In problematic cases quit the iViewETG software and restart. If this has no effect, disconnect the SMI-ETG and reboot your ETG-Laptop. Reconnect the SMI-ETG and start iViewETG. If the problem still exists contact the SMI support for further advice and help.

- During recording sessions you can close the ETG-Laptop. It will not go into sleep mode. In any other case, please avoid activating the sleep mode while the device is plugged to the USB port. Reboot your complete system if the ETG-Laptop went into sleep mode and you notice instability of the gaze cursor or the SMI-ETG does not respond correctly to the iViewETG software.

- If in any case the iViewETG software does not detect the SMI-ETG correctly (Hardware Setup 30), please disconnect the SMI-ETG, quit the iViewETG, reconnect the SMI-ETG, and restart the iViewETG software.

- When the iViewETG laptop battery has been discharged to less than 10% capacity please recharge the battery first to more than 20% capacity, before operating iViewETGagain.

7.2 Third Party Licenses

The iViewETG software uses software licensed under the *GNU Lesser General Public License (LGPL),* section 7.2.1 contains the license text:.

- *Eigen*, see http://eigen.tuxfamily.org

- *FFmpeg*, see http://ffmpeg.org/

The iViewETG software uses software licensed under the *Boost Software License*, section 7.2.2 contains the license text:

- *Boost*, see http://www.boost.org

The iViewETG software uses software licensed under the *BSD License*, section 7.2.3 contains the license text:

- *OpenCV*, see http://opencv.willowgarage.com

The iViewETG software uses the software licensed under the *GNU General Public License (GPL) version 3*, with the addition under section 7 of an exception described in the *GCC Runtime Library Exception, version 3.1*. Sections 7.2.4 and 7.2.5 contain the license text:

- libgcc and libstdc++, see http://gcc.gnu.org/onlinedocs/libstdc++

7.2.1 GNU Lesser General Public License

http://www.gnu.org/licenses/lgpl-3.0.en.html

Version 3, 29 June 2007

Copyright © 2007 Free Software Foundation, Inc. http://fsf.org/

Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

This version of the GNU Lesser General Public License incorporates the terms and conditions of version 3 of the GNU General Public License, supplemented by the additional permissions listed below.

1. **Additional Definitions.** As used herein, “this License” refers to version

3 of the GNU Lesser General Public License, and the “GNU GPL” refers to version 3 of the GNU General Public License.“The Library” refers to a covered work governed by this License, other than an Application or a Combined Work as defined below. An “Application” is any work that makes use of an interface provided by the Library, but which is not otherwise based on the Library. Defining a subclass of a class defined by the Library is deemed a mode of using an interface provided by the Library. A “Combined Work” is a work produced by combining or linking an Application with the Library. The particular version of the Library with which the Combined Work was made is also called the “Linked Version”. The “Minimal Corresponding Source” for a Combined Work means the Corresponding Source for the Combined Work, excluding any source code for portions of the Combined Work that, considered in isolation, are based on the Application, and not on the Linked Version. The “Corresponding Application Code” for a Combined Work means the object code and/or source code for the Application, including any data and utility programs needed for reproducing the Combined Work from the Application, but excluding the System Libraries of the Combined Work.

2. **Exception to Section 3 of the GNU GPL.** You may convey a covered work under sections 3 and 4 of this License without being bound by section 3 of the GNU GPL.

3. **Conveying Modified Versions.** If you modify a copy of the Library, and, in your modifications, a facility refers to a function or data to be supplied by an Application that uses the facility (other than as an argument passed when the facility is invoked), then you may convey a copy of the modified version:

   a) under this License, provided that you make a good faith effort to ensure that, in the event an Application does not supply the function or data, the facility still operates, and performs whatever part of its purpose remains meaningful, or

   b) under the GNU GPL, with none of the additional permissions of this License applicable to that copy.

 (c) 2012 SensoMotoric Instruments

4. **Object Code Incorporating Material from Library Header Files.** The object code form of an Application may incorporate material from a header file that is part of the Library. You may convey such object code under terms of your choice, provided that, if the incorporated material is not limited to numerical parameters, data structure layouts and accessors, or small macros, inline functions and templates (ten or fewer lines in length), you do both of the following:

a) Give prominent notice with each copy of the object code that the Library is used in it and that the Library and its use are covered by this License.
b) Accompany the object code with a copy of the GNU GPL and this license document.

5. **Combined Works.** You may convey a Combined Work under terms of your choice that, taken together, effectively do not restrict modification of the portions of the Library contained in the Combined Work and reverse engineering for debugging such modifications, if you also do each of the following:

a) Give prominent notice with each copy of the Combined Work that the Library is used in it and that the Library and its use are covered by this License.
b) Accompany the Combined Work with a copy of the GNU GPL and this license document.
c) For a Combined Work that displays copyright notices during execution, include the copyright notice for the Library among these notices, as well as a reference directing the user to the copies of the GNU GPL and thislicense document.
d) Do one of the following:

0) Convey the Minimal Corresponding Source under the terms of this License, and the Corresponding Application Code in a form suitable for, and under terms that permit, the user to recombine or relink the Application with a modified version of the Linked Version to produce a modified Combined Work, in

the manner specified by section 6 of the GNU GPL for conveying Corresponding Source.

1) Use a suitable shared library mechanism for linking with the Library. A suitable mechanism is one that (a) uses at run time a copy of the Library already present on the user's computer system, and (b) will operate properly with a modified version of the Library that is interface-compatible with the Linked Version.

e) Provide Installation Information, but only if you would otherwise be required to provide such information under section 6 of the GNU GPL, and only to the extent that such information is necessary to install and execute a modified version of the Combined Work produced by recombining or relinking the Application with a modified version of the Linked Version. (If you use option 4d0, the Installation Information must accompany the Minimal Corresponding Source and Corresponding Application Code. If you use option 4d1, you must provide the Installation Information in the manner specified by section 6 of the GNU GPL for conveying Corresponding Source.)

6. **Combined Libraries.** You may place library facilities that are a work based on the Library side by side in a single library together with other library facilities that are not Applications and are not covered by this License, and convey such a combined library under terms of your choice, if you do both of the following:

   a) Accompany the combined library with a copy of the same work based on the Library, uncombined with any other library facilities, conveyed under the terms of this License.
   b) Give prominent notice with the combined library that part of it is a work based on the Library, and explaining where to find the accompanying uncombined form of the same work.

7. **Revised Versions of the GNU Lesser General Public License.** The Free Software Foundation may publish revised and/or new versions of the GNU Lesser General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in

 (c) 2012 SensoMotoric Instruments

detail to address new problems or concerns.
Each version is given a distinguishing version number. If the Library as you received it specifies that a certain numbered version of the GNU Lesser General Public License “or any later version” applies to it, you have the option of following the terms and conditions either of that published version or of any later version published by the Free Software Foundation. If the Library as you received it does not specify a version number of the GNU Lesser General Public License, you may choose any version of the GNU Lesser General Public License ever published by the Free Software Foundation.
If the Library as you received it specifies that a proxy can decide whether future versions of the GNU Lesser General Public License shall apply, that proxy's public statement of acceptance of any version is permanent authorization for you to choose that version for the Library.

7.2.2 Boost Software License

http://www.boost.org/LICENSE_1_0.txt

Boost Software License - Version 1.0 - August 17th, 2003

Permission is hereby granted, free of charge, to any person or organization obtaining a copy of the software and accompanying documentation covered by this license (the "Software") to use, reproduce, display, distribute, execute, and transmit the Software, and to prepare derivative works of the Software, and to permit third-parties to whom the Software is furnished to do so, all subject to the following:

The copyright notices in the Software and this entire statement, including the above license grant, this restriction and the following disclaimer, must be included in all copies of the Software, in whole or in part, and all derivative works of the Software, unless such copies or derivative works are solely in the form of machine-executable object code generated by a source language processor.

7.2.3 BSD License

http://opensource.org/licenses/bsd-license.php

License Agreement

For Open Source Computer Vision Library
Copyright (C) 2000-2008, Intel Corporation, all rights reserved.
Copyright (C) 2008-2010, Willow Garage Inc., all rights reserved.
Third party copyrights are property of their respective owners.

Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:

- Redistribution's of source code must retain the above copyright notice, this list of conditions and the following disclaimer.
- Redistribution's in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.
- The name of the copyright holders may not be used to endorse or promote products derived from this software without specific prior written permission.

This software is provided by the copyright holders and contributors "as is" and any express or implied warranties, including, but not limited to, the implied warranties of merchantability and fitness for a particular purpose are disclaimed. In no event shall the Intel Corporation or contributors be liable for any direct, indirect, incidental, special, exemplary, or consequential damages (including, but not limited to, procurement of substitute goods or services; loss of use, data, or profits; or business interruption) however caused and on any theory of liability, whether in contract, strict liability, or tort (including negligence or otherwise) arising in any way out of the use of this software, even if advised of the possibility of such damage.

7.2.4 GNU General Public License version 3

Version 3, 29 June 2007

Copyright © 2007 Free Software Foundation, Inc. http://www.fsf.org/

Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

Preamble

The GNU General Public License is a free, copyleft license for software and other kinds of works.

The licenses for most software and other practical works are designed to take away your freedom to share and change the works. By contrast, the GNU General Public License is intended to guarantee your freedom to share and change all versions of a programm to make sure it remains free software for all its users. We, the Free Software Foundation, use the GNU General Public License for most of our software; it applies also to any other work released this way by its authors. You can apply it to your programs, too.

When we speak of free software, we are referring to freedom, not price. Our General Public Licenses are designed to make sure that you have the freedom to distribute copies of free software (and charge for them if you wish), that you receive source code or can get it if you want it, that you can change the software or use pieces of it in new free programs, and that you know you can do these things.

To protect your rights, we need to prevent others from denying you these rights or asking you to surrender the rights. Therefore, you have certain responsibilities if you distribute copies of the software, or if you modify it: responsibilities to respect the freedom of others.

For example, if you distribute copies of such a program, whether gratis or for a fee, you must pass on to the recipients the same freedoms that you received. You must make sure that they, too, receive or can get the source code. And you must show them these terms so they know their rights.

Developers that use the GNU GPL protect your rights with two steps: (1) assert copyright on the software, and (2) offer you this License giving you legal permission to copy, distribute and/or modify it.

For the developers' and authors' protection, the GPL clearly explains that there is no warranty for this free software. For both users' and authors' sake, the GPL requires that modified versions be marked as changed, so that their problems will not be attributed erroneously to authors of previous versions.

Some devices are designed to deny users access to install or run modified versions of the software inside them, although the manufacturer can do so. This is fundamentally incompatible with the aim of protecting users' freedom to change the software. The systematic pattern of such abuse occurs in the area of products for individuals to use, which is precisely where it is most unacceptable. Therefore, we have designed this version of the GPL to prohibit the practice for those products. If such problems arise substantially in other domains, we stand ready to extend this provision to those domains in future versions of the GPL, as needed to protect the freedom of users.

Finally, every program is threatened constantly by software patents. States should not allow patents to restrict development and use of software on general-purpose computers, but in those that do, we wish to avoid the special danger that patents applied to a free program could make it effectively proprietary. To prevent this, the GPL assures that patents cannot be used to render the program non-free.

The precise terms and conditions for copying, distribution and modification follow.

TERMS AND CONDITIONS

1. **Definitions.** "This License" refers to version 3 of the GNU General Public License.
   “Copyright” also means copyright-like laws that apply to other kinds of works, such as semiconductor masks.
   “The Program” refers to any copyrightable work licensed under this License. Each licensee is addressed as “you”. “Licensees” and “recipients” may be individuals or organizations.
   To “modify” a work means to copy from or adapt all or part of the work in a fashion requiring copyright permission, other than the making of an exact copy. The resulting work is called a “modified version” of the earlier work or a work “based on” the earlier work.
   A “covered work” means either the unmodified Program or a work based on the Program.
   To “propagate” a work means to do anything with it that, without permission, would make you directly or secondarily liable for infringement under applicable copyright law, except executing it on a computer or modifying a private copy. Propagation includes copying, distribution (with or without modification), making available to the public, and in some countries other activities as well.
   To “convey” a work means any kind of propagation that enables other parties to make or receive copies. Mere interaction with a user through a computer network, with no transfer of a copy, is not conveying.
   An interactive user interface displays “Appropriate Legal Notices” to the extent that it includes a convenient and prominently visible feature that (1) displays an appropriate copyright notice, and (2) tells the user that there is no warranty for the work (except to the extent that warranties are provided), that licensees may convey the work under this License, and how to view a copy of this License. If the interface presents a list of user commands or options, such as a menu, a prominent item in the list meets this criterion.

2. **Source Code.** The “source code” for a work means the preferred

form of the work for making modifications to it. “Object code” means any non-source form of a work.
A “Standard Interface” means an interface that either is an official standard defined by a recognized standards body, or, in the case of interfaces specified for a particular programming language, one that is widely used among developers working in that language.
The “System Libraries” of an executable work include anything, other than the work as a whole, that (a) is included in the normal form of packaging a Major Component, but which is not part of that Major Component, and (b) serves only to enable use of the work with that Major Component, or to implement a Standard Interface for which an implementation is available to the public in source code form. A “Major Component”, in this context, means a major essential component (kernel, window system, and so on) of the specific operating system (if any) on which the executable work runs, or a compiler used to produce the work, or an object code interpreter used to run it.
The “Corresponding Source” for a work in object code form means all the source code needed to generate, install, and (for an executable work) run the object code and to modify the work, including scripts to control those activities. However, it does not include the work’s System Libraries, or general-purpose tools or generally available free programs which are used unmodified in performing those activities but which are not part of the work. For example, Corresponding Source includes interface definition files associated with source files for the work, and the source code for shared libraries and dynamically linked subprograms that the work is specifically designed to require, such as by intimate data communication or control flow between those subprograms and other parts of the work.
The Corresponding Source need not include anything that users can regenerate automatically from other parts of the Corresponding Source.
The Corresponding Source for a work in source code form is that same work.

3. **Basic Permissions.** All rights granted under this License are

granted for the term of copyright on the Program, and are irrevocable provided the stated conditions are met. This License explicitly affirms your unlimited permission to run the unmodified Program. The output from running a covered work is covered by this License only if the output, given its content, constitutes a covered work. This License acknowledges your rights of fair use or other equivalent, as provided by copyright law.
You may make, run and propagate covered works that you do not convey, without conditions so long as your license otherwise remains in force. You may convey covered works to others for the sole purpose of having them make modifications exclusively for you, or provide you with facilities for running those works, provided that you comply with the terms of this License in conveying all material for which you do not control copyright. Those thus making or running the covered works for you must do so exclusively on your behalf, under your direction and control, on terms that prohibit them from making any copies of your copyrighted material outside their relationship with you.
Conveying under any other circumstances is permitted solely under the conditions stated below. Sublicensing is not allowed; section 10 makes it unnecessary.

4. **Protecting Users' Legal Rights From Anti-Circumvention Law.** No covered work shall be deemed part of an effective technological measure under any applicable law fulfilling obligations under article 11 of the WIPO copyright treaty adopted on 20 December 1996, or similar laws prohibiting or restricting circumvention of such measures.
When you convey a covered work, you waive any legal power to forbid circumvention of technological measures to the extent such circumvention is effected by exercising rights under this License with respect to the covered work, and you disclaim any intention to limit operation or modification of the work as a means of enforcing, against the work's users, your or third parties' legal rights to forbid circumvention of technological measures.

5. **Conveying Verbatim Copies.** You may convey verbatim copies

of the Program's source code as you receive it, in any medium, provided that you conspicuously and appropriately publish on each copy an appropriate copyright notice; keep intact all notices stating that this License and any non-permissive terms added in accord with section 7 apply to the code; keep intact all notices of the absence of any warranty; and give all recipients a copy of this License along with the Program.
You may charge any price or no price for each copy that you convey, and you may offer support or warranty protection for a fee.

6. **Conveying Modified Source Versions.** You may convey a work based on the Program, or the modifications to produce it from the Program, in the form of source code under the terms of section 4, provided that you also meet all of these conditions:

    1. The work must carry prominent notices stating that you modified it, and giving a relevant date.

    2. The work must carry prominent notices stating that it is released under this License and any conditions added under section 7. This requirement modifies the requirement in section 4 to "keep intact all notices".

    3. You must license the entire work, as a whole, under this License to anyone who comes into possession of a copy. This License will therefore apply, along with any applicable section 7 additional terms, to the whole of the work, and all its parts, regardless of how they are packaged. This License gives no permission to license the work in any other way, but it does not invalidate such permission if you have separately received it.

    4. If the work has interactive user interfaces, each must display Appropriate Legal Notices; however, if the Program has interactive interfaces that do not display Appropriate Legal Notices, your work need not make them do so.

 (c) 2012 SensoMotoric Instruments

A compilation of a covered work with other separate and independent works, which are not by their nature extensions of the covered work, and which are not combined with it such as to form a larger program, in or on a volume of a storage or distribution medium, is called an “aggregate” if the compilation and its resulting copyright are not used to limit the access or legal rights of the compilation’s users beyond what the individual works permit. Inclusion of a covered work in an aggregate does not cause this License to apply to the other parts of the aggregate.

7. **Conveying Non-Source Forms.** You may convey a covered work in object code form under the terms of sections 4 and 5, provided that you also convey the machine-readable Corresponding Source under the terms of this License, in one of these ways:

    1. Convey the object code in, or embodied in, a physical product (including a physical distribution medium), accompanied by the Corresponding Source fixed on a durable physical medium customarily used for software interchange.
    2. Convey the object code in, or embodied in, a physical product (including a physical distribution medium), accompanied by a written offer, valid for at least three years and valid for as long as you offer spare parts or customer support for that product model, to give anyone who possesses the object code either (1) a copy of the Corresponding Source for all the software in the product that is covered by this License, on a durable physical medium customarily used for software interchange, for a price no more than your reasonable cost of physically performing this conveying of source, or (2) access to copy the Corresponding Source from a network server at no charge.
    3. Convey individual copies of the object code with a copy of the written offer to provide the Corresponding Source. This alternative is allowed only occasionally and noncommercially, and only if you received the object code with such an offer, in accord with subsection 6b.

4. onvey the object code by offering access from a designated place (gratis or for a charge), and offer equivalent access to the Corresponding Source in the same way through the same place at no further charge. You need not require recipients to copy the Corresponding Source along with the object code. If the place to copy the object code is a network server, the Corresponding Source may be on a different server (operated by you or a third party) that supports equivalent copying facilities, provided you maintain clear directions next to the object code saying where to find the Corresponding Source. Regardless of what server hosts the Corresponding Source, you remain obligated to ensure that it is available for as long as needed to satisfy these requirements.
5. Convey the object code using peer-to-peer transmission, provided you inform other peers where the object code and Corresponding Source of the work are being offered to the general public at no charge under subsection 6d.

A separable portion of the object code, whose source code is excluded from the Corresponding Source as a System Library, need not be included in conveying the object code work.
A “User Product” is either (1) a “consumer product”, which means any tangible personal property which is normally used for personal, family, or household purposes, or (2) anything designed or sold for incorporation into a dwelling. In determining whether a product is a consumer product, doubtful cases shall be resolved in favor of coverage. For a particular product received by a particular user, “normally used” refers to a typical or common use of that class of product, regardless of the status of the particular user or of the way in which the particular user actually uses, or expects or is expected to use, the product. A product is a consumer product regardless of whether the product has substantial commercial, industrial or non-consumer uses, unless such uses represent the only significant mode of use of the product.
“Installation Information” for a User Product means any methods, procedures, authorization keys, or other information required to install and execute modified versions of a covered work in that User Product from a

modified version of its Corresponding Source. The information must suffice to ensure that the continued functioning of the modified object code is in no case prevented or interfered with solely because modification has been made.
If you convey an object code work under this section in, or with, or specifically for use in, a User Product, and the conveying occurs as part of a transaction in which the right of possession and use of the User Product is transferred to the recipient in perpetuity or for a fixed term (regardless of how the transaction is characterized), the Corresponding Source conveyed under this section must be accompanied by the Installation Information. But this requirement does not apply if neither you nor any third party retains the ability to install modified object code on the User Product (for example, the work has been installed in ROM).
The requirement to provide Installation Information does not include a requirement to continue to provide support service, warranty, or updates for a work that has been modified or installed by the recipient, or for the User Product in which it has been modified or installed. Access to a network may be denied when the modification itself materially and adversely affects the operation of the network or violates the rules and protocols for communication across the network.
Corresponding Source conveyed, and Installation Information provided, in accord with this section must be in a format that is publicly documented (and with an implementation available to the public in source code form), and must require no special password or key for unpacking, reading or copying.

8. **Additional Terms.** "Additional permissions" are terms that supplement the terms of this License by making exceptions from one or more of its conditions. Additional permissions that are applicable to the entire Program shall be treated as though they were included in this License, to the extent that they are valid under applicable law. If additional permissions apply only to part of the Program, that part may be used separately under those permissions, but the entire Program remains governed by this License without regard to the additional permissions.
When you convey a copy of a covered work, you may at your option remove any additional permissions from that copy, or from

any part of it. (Additional permissions may be written to require their own removal in certain cases when you modify the work.) You may place additional permissions on material, added by you to a covered work, for which you have or can give appropriate copyright permission.
Notwithstanding any other provision of this License, for material you add to a covered work, you may (if authorized by the copyright holders of that material) supplement the terms of this License with terms:

1. Disclaiming warranty or limiting liability differently from the terms of sections 15 and 16 of this License; or
2. Requiring preservation of specified reasonable legal notices or author attributions in that material or in the Appropriate Legal Notices displayed by works containing it; or
3. Prohibiting misrepresentation of the origin of that material, or requiring that modified versions of such material be marked in reasonable ways as different from the original version; or
4. Limiting the use for publicity purposes of names of licensors or authors of the material; or
5. Declining to grant rights under trademark law for use of some trade names, trademarks, or service marks; or
6. Requiring indemnification of licensors and authors of that material by anyone who conveys the material (or modified versions of it) with contractual assumptions of liability to the recipient, for any liability that these contractual assumptions directly impose on those licensors and authors.

All other non-permissive additional terms are considered “further restrictions” within the meaning of section 10. If the Program as you received it, or any part of it, contains a notice stating that it is governed by this License along with a term that is a further restriction, you may remove that term. If a license document contains a further restriction but permits relicensing or conveying under this License, you may add to a covered work material governed by the terms of that license document, provided that the further restriction does not survive such relicensing or conveying. If you add terms to a covered work in accord with this section, you must

place, in the relevant source files, a statement of the additional terms that apply to those files, or a notice indicating where to find the applicable terms.
Additional terms, permissive or non-permissive, may be stated in the form of a separately written license, or stated as exceptions; the above requirements apply either way.

9. **Termination.** You may not propagate or modify a covered work except as expressly provided under this License. Any attempt otherwise to propagate or modify it is void, and will automatically terminate your rights under this License (including any patent licenses granted under the third paragraph of section 11).
   However, if you cease all violation of this License, then your license from a particular copyright holder is reinstated (a) provisionally, unless and until the copyright holder explicitly and finally terminates your license, and (b) permanently, if the copyright holder fails to notify you of the violation by some reasonable means prior to 60 days after the cessation.
   Moreover, your license from a particular copyright holder is reinstated permanently if the copyright holder notifies you of the violation by some reasonable means, this is the first time you have received notice of violation of this License (for any work) from that copyright holder, and you cure the violation prior to 30 days after your receipt of the notice.
   Termination of your rights under this section does not terminate the licenses of parties who have received copies or rights from you under this License. If your rights have been terminated and not permanently reinstated, you do not qualify to receive new licenses for the same material under section.

10. **Acceptance Not Required for Having Copies.** You are not required to accept this License in order to receive or run a copy of the Program. Ancillary propagation of a covered work occurring solely as a consequence of using peer-to-peer transmission to receive a copy likewise does not require acceptance. However, nothing other than this License grants you permission to propagate or modify any covered work. These actions infringe copyright if you

do not accept this License. Therefore, by modifying or propagating a covered work, you indicate your acceptance of this License to do so.

11. **Automatic Licensing of Downstream Recipients.** Each time you convey a covered work, the recipient automatically receives a license from the original licensors, to run, modify and propagate that work, subject to this License. You are not responsible for enforcing compliance by third parties with this License.
An “entity transaction” is a transaction transferring control of an organization, or substantially all assets of one, or subdividing an organization, or merging organizations. If propagation of a covered work results from an entity transaction, each party to that transaction who receives a copy of the work also receives whatever licenses to the work the party’s predecessor in interest had or could give under the previous paragraph, plus a right to possession of the Corresponding Source of the work from the predecessor in interest, if the predecessor has it or can get it with reasonable efforts.
You may not impose any further restrictions on the exercise of the rights granted or affirmed under this License. For example, you may not impose a license fee, royalty, or other charge for exercise of rights granted under this License, and you may not initiate litigation (including a cross-claim or counterclaim in a lawsuit) alleging that any patent claim is infringed by making, using, selling, offering for sale, or importing the Program or any portion of it.

12. **Patents.** A “contributor” is a copyright holder who authorizes use under this License of the Program or a work on which the Program is based. The work thus licensed is called the contributor’s “contributor version”.
A contributor’s “essential patent claims” are all patent claims owned or controlled by the contributor, whether already acquired or hereafter acquired, that would be infringed by some manner, permitted by this License, of making, using, or selling its contributor version, but do not include claims that would be infringed only as a consequence of further modification of the contributor version. For purposes of this definition, “control” includes

the right to grant patent sublicenses in a manner consistent with the requirements of this License.
Each contributor grants you a non-exclusive, worldwide, royalty-free patent license under the contributor's essential patent claims, to make, use, sell, offer for sale, import and otherwise run, modify and propagate the contents of its contributor version.
In the following three paragraphs, a "patent license" is any express agreement or commitment, however denominated, not to enforce a patent (such as an express permission to practice a patent or covenant not to sue for patent infringement). To "grant" such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party.
If you convey a covered work, knowingly relying on a patent license, and the Corresponding Source of the work is not available for anyone to copy, free of charge and under the terms of this License, through a publicly available network server or other readily accessible means, then you must either (1) cause the Corresponding Source to be so available, or (2) arrange to deprive yourself of the benefit of the patent license for this particular work, or (3) arrange, in a manner consistent with the requirements of this License, to extend the patent license to downstream recipients. "Knowingly relying" means you have actual knowledge that, but for the patent license, your conveying the covered work in a country, or your recipient's use of the covered work in a country, would infringe one or more identifiable patents in that country that you have reason to believe are valid.
If, pursuant to or in connection with a single transaction or arrangement, you convey, or propagate by procuring conveyance of, a covered work, and grant a patent license to some of the parties receiving the covered work authorizing them to use, propagate, modify or convey a specific copy of the covered work, then the patent license you grant is automatically extended to all recipients of the covered work and works based on it.
A patent license is "discriminatory" if it does not include within the scope of its coverage, prohibits the exercise of, or is conditioned on the non-exercise of one or more of the rights that are specifically granted under this License. You may not convey a covered work if

you are a party to an arrangement with a third party that is in the business of distributing software, under which you make payment to the third party based on the extent of your activity of conveying the work, and under which the third party grants, to any of the parties who would receive the covered work from you, a discriminatory patent license (a) in connection with copies of the covered work conveyed by you (or copies made from those copies), or (b) primarily for and in connection with specific products or compilations that contain the covered work, unless you entered into that arrangement, or that patent license was granted, prior to 28 March 2007.
Nothing in this License shall be construed as excluding or limiting any implied license or other defenses to infringement that may otherwise be available to you under applicable patent law.

13. **No Surrender of Others' Freedom.** If conditions are imposed on you (whether by court order, agreement or otherwise) that contradict the conditions of this License, they do not excuse you from the conditions of this License. If you cannot convey a covered work so as to satisfy simultaneously your obligations under this License and any other pertinent obligations, then as a consequence you may not convey it at all. For example, if you agree to terms that obligate you to collect a royalty for further conveying from those to whom you convey the Program, the only way you could satisfy both those terms and this License would be to refrain entirely from conveying the Program.

14. **Use with the GNU Affero Gnereral Public License.**
Notwithstanding any other provision of this License, you have permission to link or combine any covered work with a work licensed under version 3 of the GNU Affero General Public License into a single combined work, and to convey the resulting work. The terms of this License will continue to apply to the part which is the covered work, but the special requirements of the GNU Affero General Public License, section 13, concerning interaction through a network will apply to the combination as such.

15. **Revised Versions of this License.** The Free Software Foundation may publish revised and/or new versions of the GNU General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.
    Each version is given a distinguishing version number. If the Program specifies that a certain numbered version of the GNU General Public License “or any later version” applies to it, you have the option of following the terms and conditions either of that numbered version or of any later version published by the Free Software Foundation. If the Program does not specify a version number of the GNU General Public License, you may choose any version ever published by the Free Software Foundation.
    If the Program specifies that a proxy can decide which future versions of the GNU General Public License can be used, that proxy’s public statement of acceptance of a version permanently authorizes you to choose that version for the Program.
    Later license versions may give you additional or different permissions. However, no additional obligations are imposed on any author or copyright holder as a result of your choosing to follow a later version.

16. **Disclaimer of Warranty.** THERE IS NO WARRANTY FOR THE PROGRAM, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE PROGRAM “AS IS” WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM IS WITH YOU. SHOULD THE PROGRAM PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION.

17. **Limitation of Liability.** IN NO EVENT UNLESS REQUIRED BY APPLICABLE LAW OR AGREED TO IN WRITING WILL ANY

COPYRIGHT HOLDER, OR ANY OTHER PARTY WHO MODIFIES AND/OR CONVEYS THE PROGRAM AS PERMITTED ABOVE, BE LIABLE TO YOU FOR DAMAGES, INCLUDING ANY GENERAL, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE USE OR INABILITY TO USE THE PROGRAM (INCLUDING BUT NOT LIMITED TO LOSS OF DATA OR DATA BEING RENDERED INACCURATE OR LOSSES SUSTAINED BY YOU OR THIRD PARTIES OR A FAILURE OF THE PROGRAM TO OPERATE WITH ANY OTHER PROGRAMS), EVEN IF SUCH HOLDER OR OTHER PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

18. **Interpretation of Sections 15 and 16.**
If the disclaimer of warranty and limitation of liability provided above cannot be given local legal effect according to their terms, reviewing courts shall apply local law that most closely approximates an absolute waiver of all civil liability in connection with the Program, unless a warranty or assumption of liability accompanies a copy of the Program in return for a fee.

7.2.5 GCC Runtime Library Exception

http://gcc.gnu.org/onlinedocs/libstdc++/manual/license.html

GCC RUNTIME LIBRARY EXCEPTION

Version 3.1, 31 March 2009

Copyright (C) 2009 Free Software Foundation, Inc.

Everyone is permitted to copy and distribute verbatim copies of this license document, but changing it is not allowed.

This GCC Runtime Library Exception ("Exception") is an additional permission under section 7 of the GNU General Public License, version 3 ("GPLv3"). It applies to a given file (the "Runtime Library") that bears a

notice placed by the copyright holder of the file stating that the file is governed by GPLv3 along with this Exception.

When you use GCC to compile a program, GCC may combine portions of certain GCC header files and runtime libraries with the compiled program. The purpose of this Exception is to allow compilation of non-GPL (including proprietary) programs to use, in this way, the header files and runtime libraries covered by this Exception.

1. **Definitions.** A file is an "Independent Module" if it either requires the Runtime Library for execution after a Compilation Process, or makes use of an interface provided by the Runtime Library, but is not otherwise based on the Runtime Library.
   "GCC" means a version of the GNU Compiler Collection, with or without modifications, governed by version 3 (or a specified later version) of the GNU General Public License (GPL) with the option of using any subsequent versions published by the FSF.
   "GPL-compatible Software" is software whose conditions of propagation, modification and use would permit combination with GCC in accord with the license of GCC.
   "Target Code" refers to output from any compiler for a real or virtual target processor architecture, in executable form or suitable for input to an assembler, loader, linker and/or execution phase. Notwithstanding that, Target Code does not include data in any format that is used as a compiler intermediate representation, or used for producing a compiler intermediate representation.
   The "Compilation Process" transforms code entirely represented in non-intermediate languages designed for human-written code, and/or in Java Virtual Machine byte code, into Target Code. Thus, for example, use of source code generators and preprocessors need not be considered part of the Compilation Process, since the Compilation Process can be understood as starting with the output of the generators or preprocessors.
   A Compilation Process is "Eligible" if it is done using GCC, alone or with other GPL-compatible software, or if it is done without using any work based on GCC. For example, using non-GPL-compatible Software to optimize any GCC intermediate representations would not qualify as an

Eligible Compilation Process.

2. **Grant of Additional Permission.** You have permission to propagate a work of Target Code formed by combining the Runtime Library with Independent Modules, even if such propagation would otherwise violate the terms of GPLv3, provided that all Target Code was generated by Eligible Compilation Processes. You may then convey such a combination under terms of your choice, consistent with the licensing of the Independent Modules.

3. **No Weakening of GCC Copyleft.** No Weakening of GCC Copyleft. The availability of this Exception does not imply any general presumption that third-party software is unaffected by the copyleft requirements of the license of GCC.

Hopefully that text is self-explanatory. If it isn't, you need to speak to your lawyer, or the Free Software Foundation.

 (c) 2012 SensoMotoric Instruments

License Agreement and Warranty

Chapter



8 License Agreement and Warranty

IMPORTANT – PLEASE READ CAREFULLY: This license agreement ("Agreement") is an agreement between you (either an individual or a company, "Licensee") and SensoMotoric Instruments GmbH ("SMI"). The "Licensed Materials" provided to Licensee subject to this Agreement include the software programs, which were installed on the equipment Licensee purchased from SMI (the "Designated Equipment") at the time of delivery of the Designated Equipment and/or other software programs SMI has granted Licensee access to (the "Software") as well as any "on-line" or electronic documentation associated with the Software, or any portion thereof (the "Documentation"), as well as any updates or upgrades to the Software and Documentation, if any, or any portion thereof, provided to Licensee at SMI's sole discretion.
By installing, downloading, copying or otherwise using the Licensed Materials, you agree to abide by the following provisions. This Agreement is displayed for you to read prior to using the Licensed Materials.
The Software is provided with and uses libraries from the FFmpeg project under the LGPLv2.1, GNU Lesser General Public License (LGPL) version 2.1 or later (the "FFmpeg Libraries"), which are not owned by SMI. The FFmpeg code owners' contact information and the LGPLv2.1 licensing terms and conditions can be found at http://ffmpeg.org
If you do not agree with these provisions, do not download, install or use the Licensed Materials. If you have already paid for the Licensed Materials, you may return them for a full refund to: SensoMotoric Instruments, Warthestraße 21, 14513 Teltow, Germany. If the Licensed Materials were installed on Designated Equipment, you may also return such Designated Equipment for a full refund.

1. **License.** Subject to the terms of this Agreement, SMI hereby grants and Licensee accepts a non-transferable, non-exclusive, non-assignable license without the right to sublicense to use the Licensed Materials only (i) for Licensee's business operations, (ii) on the Designated Equipment, if any, or on a back-up equipment while the Designated

Equipment is inoperable, and (iii) in accordance with the Documentation. Licensee may make one copy of the Software in machine readable form for backup purposes only; every notice on the original will be replicated on the copy. Installation of the Software, if any, is Licensee's sole responsibility.

2. **Rights in Licensed Materials.** Title to and ownership in the Licensed Materials and all proprietary rights with respect to the Licensed Materials and all copies and portions thereof, remain exclusively with SMI. The Agreement does not constitute a sale of the Licensed Materials or any portion or copy of it. Title to and ownership in Licensee's application software that makes calls to but does not contain all or any portion of the Software remains with Licensee, but such application software may not be licensed or otherwise transferred to third parties without SMI's prior written consent.

3. **Confidentiality.** Licensed Materials are proprietary to SMI and constitute SMI trade secrets. Licensee shall maintain Licensed Materials in confidence and prevent their disclosure using at least the same degree of care it uses for its own trade secrets, but in no event less than a reasonable degree of care. Licensee shall not disclose Licensed Materials or any part thereof to anyone for any purpose, other than to its employees and sub-contractors for the purpose of exercising the rights expressly granted under this Agreement, provided they have in writing agreed to confidentiality obligations at least equivalent to the obligations stated herein.

4. **No Reverse Engineering**. License shall not, and shall not allow any third party to, decompile, disassemble or otherwise reverse engineer or by any means whatsoever attempt to reconstruct or discover any source code or underlying ideas, algorithms, file formats or programming or interoperability interfaces of the Software or of any files contained or generated using the Software. The obligations under this Section shall survive any termination of the Agreement. Licensee may apply to SMI at the address above for further information on the interface between the Software and other applications; SMI will on reasonable terms and conditions provide such information as the Licensee may reasonably

require to enable the Software to interoperate with its applications. These restrictions do not apply to the FFmpeg Libraries.

5. Warranty.

a)If Licensed Materials show a defect during the limitation period specified in Sec. 5 d), which already existed at the time of transfer of risk, SMI will provide Licensee with a corrected version of such defective Licensed Material ("Repeat Performance").

b)Licensed Materials are defective, if they do not conform to the written agreement between SMI and Licensee, if any, or to SMI's specification.

c)Licensee's claims for defects are excluded:

- for insignificant divergences from the agreement or specification;
- for damages caused after the transfer of risk by external influences such as fire, water, currency surge, etc.; improper installation, operation, use or maintenance; use in fields of application and environmental conditions other than those expressly specified by SMI; and use in combination with other products not approved by SMI for this purpose, excessive stress or normal wear and tear;
- for defects of the Software, which cannot be reproduced;
- for Licensee's or a third party's modifications of Licensed Materials and results therefrom;
- in as far as Licensee has not notified SMI of apparent defects within 5 business days of delivery and of hidden defects within 5 business days of discovery.

d)The limitation period for warranty claims is 12 months. This does not apply in cases of fraud, intent, gross negligence and personal injury. Repeat performance does not restart the limitation period.

e)Licensee has to always first provide SMI the opportunity for Repeat Performance within a reasonable period of time. Repeat Performance does not constitute acceptance of a legal obligation.

f)If the Repeat Performance fails even within an additional period of time of reasonable length to be granted by Licensee, Licensee shall, notwithstanding its claims under Section 6, if any, be entitled to rescind the contract or to claim a price reduction.

 (c) 2012 SensoMotoric Instruments

g)If the analysis of an alleged defect shows that it is not covered by the above warranty, SMI may charge for the failure analysis and correction of the defect, if any, at SMI's then applicable rates.

6. Liability Limitations.

a)SMI is only liable for damages caused by slight negligence if such are due to a material breach of duty, which endangers the achievement of the objective of the Agreement, or the failure to comply with duties, the very discharge of which is an essential prerequisite for the proper performance of the Agreement.

b)In cases of Section 6 a), the liability is limited to the damage, which is typical for contracts like this Agreement and which could have been foreseen.

c)SMI's liability is also limited to the damage, which is typical for contracts like this Agreement and which could have been foreseen for damages caused by the gross negligence of an agent or an employee of SMI, who is not an officer or executive of SMI.

d)In cases of Sections 6 a) and c), SMI's liability is limited to a maximum amount of EURO 500.000, respectively EURO 100.000 for financial losses.

e)Licensee's claims for damages caused by slight negligence or caused by the gross negligence of an agent or an employee of SMI, who is not an officer or executive of SMI, which are not based on defects and thus not subject to time-barring in accordance with Sec. 5 d) are time-barred at the latest 2 years from the point in time Licensee became aware of the damage and regardless of Licensee's awareness at the latest 3 years after the damaging event.

f)With the exception of liability under the Product Liability Law, for defects after having given a guarantee, for fraudulently concealed defects and for personal injury, the above limitations of liability shall apply to all claims, irrespective of their legal basis, in particular to all claims based on breach of contract or tort.

g)The above limitations of liability also apply in case of Licensee's claims for damages against SMI's employees or agents.

7. **SMI Indemnity.** SMI will defend or settle any action brought against Licensee to the extent based on a claim that Licensed Materials, used

within the scope of the license granted in this Agreement, infringe any copyright valid in the European Union or the European Economic Area and will pay the cost of any final settlement or judgment attributable to such claim, provided (i) Licensee has given prompt notice to SMI of such claim, (ii) Licensee has not recognized an infringement of the third party's copyright, and (iii) the entire defense and the settlement negotiations are reserved for SMI with Licensee's full cooperation and support. If Licensee discontinues the use of Licensed Materials for damage reduction or other important reasons, Licensee is obligated to point out to the third party that the discontinuation of use does not constitute an acknowledgement of a copyright infringement. If SMI believes Licensed Materials are likely to be the subject of an infringement claim, it may elect to replace or modify such Software or Documentation to make it non-infringing or terminate the Agreement on written notice to Licensee. SMI shall have no obligation to defend (or any other liability) to the extent any claim involves a Software release other than the current, unaltered release, if such would have avoided infringement, or use of the Software in combination with non-SMI programs or data, unless the infringement would also incur without such combination. The foregoing states the entire obligation and liability of SMI with respect to any infringement by Licensed Materials of any intellectual property rights or other proprietary rights of Licensee or a third party without prejudice to any claims for damages in accordance with Section 6.

8. **Licensee Indemnity.** Licensee will defend and indemnify SMI, and hold it harmless from all costs, including attorney's fees, arising from any claim that may be made against SMI by any third party as a result of Licensee's use of Licensed Materials, excluding claims for which SMI is obligated to defend or indemnify Licensee under Section 7.

9. **Export Restriction.** Licensee will not remove or export from Germany or from the country Licensed Materials were originally shipped to by SMI or re-export from anywhere any part of the Licensed Materials or any direct product of the Software except in compliance with all applicable export laws and regulations, including without limitation, those of the U. S. Department of Commerce.

 (c) 2012 SensoMotoric Instruments

10. **Non-Waiver; Severability; Non-Assignment.** The delay or failure of either party to exercise any right provided in this Agreement shall not be deemed a waiver. If any provision of this Agreement is held invalid, all others shall remain in force. Licensee may not, in whole or in part, assign or otherwise transfer this Agreement or any of its rights or obligations hereunder.

11. **Termination.** This Agreement may be terminated without any fee reduction (i) by Licensee without cause on 30 days notice; (ii) by SMI, in addition to other remedies, if Licensee fails to cure any breach of its obligations hereunder within 30 days of notice thereof; (iii) on notice by either party if the other party ceases to do business in the normal course, becomes insolvent, or becomes subject to any bankruptcy, insolvency, or equivalent proceedings. Upon termination by either party for any reason, Licensee shall at SMI's instructions immediately destroy or return the Licensed Materials and all copies thereof to SMI and delete the Software and all copies thereof from the Designated Equipment.

12. **Entire Agreement; Written Form Requirement.** There are no separate oral agreements; any supplementary agreements or modifications hereto must be made in writing. This also applies to any waiver of this requirement of written form.

13. **Notices.** All notices under the Agreement must be in writing and shall be delivered by hand or by overnight courier to the addresses of the parties set forth above.

14. **Applicable Law and Jurisdiction.** German law applies with the exception of its conflict of laws rules. The application of the United Nations Convention on Contracts for the International Sale of Goods (CISG) is expressly excluded The courts of Berlin, Germany, shall have exclusive jurisdiction for any action brought under or in connection with this Agreement.

Teltow, Germany, 2004- 2011
SensoMotoric Instruments GmbH

About SMI

Chapter



9 About SMI

SensoMotoric Instruments (SMI) is a world leader in dedicated computer vision applications, developing and marketing eye & gaze tracking systems and OEM solutions for a wide range of applications.

Founded in 1991 as a spin-off from academic research, SMI was the first company to offer a commercial, vision-based 3D eye tracking solution. We now have over 20 years of experience in developing application-specific solutions in close collaboration with our clients.

We serve our customers around the globe from our offices in Teltow, near Berlin, Germany and Boston, USA, backed by a network of trusted local partners in many countries.

Our products combine a maximum of performance and usability with the highest possible quality, resulting in high-value solutions for our customers. Our major fields of expertise are:

- Eye & gaze tracking systems in research and industry
- High speed image processing, and
- Eye tracking and registration solutions in ophthalmology.

More than 5,000 of our systems installed worldwide are testimony to our continuing success in providing innovative products and outstanding services to the market. While SMI has won several awards, the largest reward for us each year is our trusted business relationships with academia and industry.

Please contact us:

International Headquarters

SensoMotoric Instruments GmbH (SMI)
Warthestraße 21
D-14513 Teltow/Berlin
GERMANY

 (c) 2012 SensoMotoric Instruments

Phone: +49 (3328) 3955 0
Fax: +49 (3328) 3955 99
e-mail: info@smi.de

North American Headquarters

SensoMotoric Instruments, Inc.
28 Atlantic Avenue
236 Lewis Wharf
Boston, MA 02110
USA

Phone: +1 - 617 - 557 - 0010
Fax: +1 - 617 - 507 - 83 19
Toll-Free: 888 SMI USA1
e-mail: info@smivision.com

Please also visit our home page:

http://www.smivision.com

Copyright (c) 2012 SensoMotoric Instruments
Last updated: 12. April 2012

Index

- 0 -

0-PointCalibration 65

- 1 -

1-PointCalibration 67

- 3 -

3-PointCalibration 68

- A -

About SMI 124

- B -

Battery Life 77

- C -

Calibration 64
Calibration Reset 71
Contact Lenses 74

- D -

Dashboard 75
Data Storage Structure 83
Declaration of Conformity 8
Directory Structure 86

- E -

Experiment
  Close Experiment 45, 48
  New Experiment 46
  Open Experiment 47
ExperimentSettings 53
Eye Position 73

- F -

FlightCase 16
Fluorescent 56
Fluorescent High 57

- G -

Gaze Cursor Stabilization 72

- H -

Hardware Configuration 55
Head Strap 13

- I -

IR Reflections 74
iView ETG Software 36

- L -

Lenses 91
License Agreement 116

- M -

Maintenance 7
Make Up 73
Microphone 42
Monocular Vision 73

- N -

Nose Rests 32

(c) 2012 SensoMotoric Instruments

- Q -

Quick Run 25
Quick Start 24

- R -

Recording Display 78
Recording Screen 77
Respondent 62

- S -

Settings 49

- T -

Technical Specifications 19
Trackability 78
Tungsten 57

- U -

UserInterface 75

- W -

White Balancing 56

(c) 2012 SensoMotoric Instruments

